Artificial Intelligence for Engineering Design, Analysis and Manufacturing

www.cambridge.org/aie

Creation-as-transmission: a cognitive-based framework for cultural heritage learning through AI collaborative creation

Yin Cui[1], ShiJun Ge[2], Junfeng Wang[3] and Xiaolong Wang[3]

[1]School of Design and Inovation, Shenzhen Technology University, Shenzhen, China; [2]Department of Art Design, Beijing City University, Beijing, China and [3]Shenzhen Technology University, Shenzhen, China

Research Article

Cite this article: Cui Y, Ge S, Wang J and Wang X (2026). Creation-as-transmission: a cognitive-based framework for cultural heritage learning through AI collaborative creation. *Artificial Intelligence for Engineering Design, Analysis and Manufacturing*, **40**, e5, 1–26
https://doi.org/10.1017/S0890060426100237

Received: 18 August 2025
Revised: 07 January 2026
Accepted: 16 January 2026

Keywords:
cultural heritage digitization; AI collaborative creation; participatory design; human–AI collaboration; multimodal interaction; intangible cultural heritage

Corresponding author:
Junfeng Wang;
Email: wangjunfeng@sztu.edu.cn

Abstract

Digital transmission of traditional cultural heritage faces challenges of insufficient user engagement and superficial dissemination. Using the Cizhou kiln as a case study, this research proposes the "Creation-as-Transmission" design paradigm, which transcends traditional unidirectional display models by transforming users from cultural consumers to cultural co-creators. This paradigm systematically integrates situated cognition theory and co-creation theory, achieving deep cultural learning and socialized transmission through AI collaborative creation. Empirical studies ($N = 200$) demonstrate that compared to traditional methods, this paradigm significantly enhances cultural cognition (74% improvement) and the proportion of spontaneous sharing on social media (82.6% sharing rate). More importantly, this research provides the first quantification of causal mechanisms between creative behaviors and cultural cognition, establishing a predictive model explaining 73.1% of cognitive variance and offering mechanistic evidence for situated learning theory's application in cultural heritage transmission. The research contributions have threefold value: theoretically, it expands HCI understanding of the relationship between "design behaviors and cognitive effects"; methodologically, it provides a transferable implementation framework for other intangible cultural heritage; practically, it opens new pathways for digital technologies to promote sustainable cultural transmission.

Introduction: research background, challenges, and objectives

The digitization of cultural heritage has become a global issue, with UNESCO listing the application of digital technologies to cultural heritage protection as a key strategy (UNESCO, 2022). However, the transmission of traditional culture in the digital age faces challenges such as insufficient user participation and superficial content delivery (Economou, 2023). This research takes Chinese Cizhou kiln culture as an example to explore a new "design driven + technology supported" cultural transmission method, achieving a paradigm shift from passive acceptance to active participation through AI collaborative creation.

Research background and challenges

As the largest folk kiln system in northern China, the Cizhou kiln, with its decorative patterns characterized by black–and–white contrast and life–centered artistic features, has become an important carrier of traditional Chinese culture and was listed in the first batch of national intangible cultural heritage in 2006 (Qin, 2024). Our in-depth survey of 481 Chinese youth revealed three major challenges facing Cizhou kiln cultural transmission:

1. Cognitive gap: 43% of respondents indicated they had never been exposed to Cizhou kiln culture, reflecting the transmission difficulties of traditional culture among the younger generation;
2. Transmission bottleneck: Analysis of different transmission methods revealed that traditional exhibitions and video approaches achieved user retention rates of only 43.62% and 52.38%, respectively, indicating that unidirectional transmission models struggle to maintain sustained user interest (Feng et al., 2022);
3. Participation barriers: Cizhou kiln pattern creation requires years of professional training (Jin, 2018), and such high-threshold skill requirements significantly limit public participation possibilities.

Related research progress and gaps

In recent years, there has been continuous innovation in cultural heritage digitization practices. In interactive experience design, research shows that compared to traditional "learning-based" approaches, "experiential" participation demonstrates stronger cultural transmission efficacy (Rather, 2020; Alsaqqaf and Li, 2023; Yu et al., 2024). Co-creation, as an emerging design

© The Author(s), 2026. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0), which permits unrestricted re-use, distribution and reproduction, provided the original article is properly cited.

paradigm, significantly enhances engagement and satisfaction by promoting deep interaction between users and technology (Kohler et al., 2011; Avni et al., 2025). In technological applications, Artificial Intelligence Generated Content (AIGC) technology has achieved significant progress in cultural heritage protection, such as Dunhuang mural restoration (Yang, 2024) and brocade pattern generation (Sun et al., 2024). Despite these studies demonstrating the application potential of digital technologies, key challenges remain: (1) Limitations in participation methods: Traditional unidirectional display methods struggle to inspire user engagement (Liu, 2020; Kong, 2021); (2) Technology-culture adaptation issues: Existing AIGC systems support creative generation but lack deep modeling of specific cultural heritage styles, leading to deviations between generated content and historical characteristics (Sun et al., 2019; Zhang et al., 2022); (3) Incomplete evaluation systems: Current research primarily focuses on technical implementation, lacking systematic verification of user participation experience and cultural cognitive enhancement effects (Li et al., 2021).

In summary, despite significant progress in cultural heritage digitization across various dimensions, existing approaches have largely failed to systematically address the core challenge of integrating "deep cultural learning," "user-driven creation," and "socialized transmission." Specifically, traditional digital displays (such as virtual museums) focus on unidirectional information delivery, positioning users in passive roles; interactive experience systems (such as AR/VR tours) enhance immersion but still cast users primarily as content consumers rather than creators; while the few AIGC applications that incorporate creative elements often struggle to balance cultural fidelity with user-friendliness, or lack closed-loop designs that effectively transform individual creative behaviors into cultural transmission momentum. The "Creation-as-Transmission" paradigm proposed in this research aims to bridge this comprehensive gap: by integrating situated cognition theory and co-creation theory, it constructs an interaction framework that positions users at the core of cultural creation and transmission. This framework not only achieves technical balance between high cultural fidelity and low creation barriers, but also establishes a transmission loop from individual creation to social sharing at the mechanistic level, thereby fundamentally distinguishing itself from and transcending existing cultural heritage digitization models, demonstrating its unique theoretical contribution and practical innovation value.

Research questions

Given the above challenges, this research proposes the following core research questions:

1. How can we design an AI collaborative creation framework that effectively maintains cultural characteristic authenticity while lowering cultural creation barriers?
2. Does the AI collaborative creation paradigm have significant advantages over traditional unidirectional transmission methods in terms of cultural knowledge acquisition and participation motivation?
3. How do users perceive and evaluate the collaborative experience during interaction with AI collaborative creationsystems?

Research contributions

This research proposes an innovative "design-driven + technology-supported" solution with the following main contributions:

1. Theoretical contribution: This study proposes the "Creation-as-Transmission" design paradigm. This approach transforms users from passive consumers into active creators. It also broadens how we understand cultural learning in human-computer interaction by showing that creative activities play a unique role as bridges in cultural learning. This framework provides new theoretical foundations for HCI cultural heritage interaction design through systematic integration of situated cognition and co-creation theories.
2. Technical implementation: Constructs a professional Cizhou kiln database containing 6,000 high-quality annotated samples and integrates a multimodal generation framework based on Stable Diffusion-XL and IP-Adapter. The proposed VAE(Variational Autoencoder) + LoRA(Low-Rank Adaptation)-based multi-scale style transfer algorithm significantly improves generation quality (FID = 21.7, PSNR = 38.6 dB) while maintaining high cultural characteristic fidelity (89.4%).
3. Empirical validation: Through multi-method research ($N = 200$), the study quantitatively demonstrates the significant effects of the AI collaborative creation system in enhancing cultural cognition (accuracy 94% vs 54%, $p < 0.001$) and cultural interest levels, providing a replicable and transferable practical framework for cultural heritage interaction design.

Furthermore, through deep mining of system log data, this research provides the first quantification of causal pathways between creative behaviors (iteration frequency, element exploration, interaction patterns) and cultural cognitive enhancement, establishing a predictive model that explains 73.1% of cognitive variance. This provides mechanistic evidence for the application of situated cognition theory in cultural heritage transmission and enriches HCI understanding of the "design behavior-cognitive effect" relationship.

Theoretical framework and related research progress

Artistic characteristics of Cizhou Kiln

As a representative folk kiln system in northern China, the Cizhou kiln is renowned for its diverse products including daily utensils, household items, and cultural artifacts. Typical wares of the Cizhou Kiln are shown in Table 1. Through field investigations and literature research (Tian, 1985; Zhang, 2000; Shang, 2015), this study systematically analyzes the characteristics of Cizhou kiln decorative patterns from the Song, Jin, and Yuan periods:

1. Morphological characteristics: Simple and elegant, reflecting the rustic nature of folk kilns and Song dynasty Neo-Confucian aesthetics (Ma, 2000); regular vessel forms with coordinated proportions; emphasis on practicality to meet daily needs.
2. Craft characteristics: Primarily featuring black-and-white contrast as the main expressive technique, with achromatic color schemes displaying restrained and elegant aesthetics, derived from local, unique porcelain clay properties (Qin, 2024). Compositions are divided into symmetrical types (such as peony patterns, dignified and stable) and asymmetrical types (such as children-at-play patterns, lively and dynamic); featuring a unique "three-seven composition" principle, with subject-to-background area ratio of approximately 3:7.

These characteristics form the unique artistic language of the Cizhou kiln and represent the core cultural elements that must be preserved in this research system design.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

Table 1. Typical wares of the Cizhou Kiln

Children's Fishing Pillow of Cizhou Kiln (Song Dynasty) (Collected by Hebei Provincial Museum)	Infant play pattern bean-shaped pillow (Jin Dynasty) (Collected by Taihang Ancient Ceramics Art Museum)	Octagonal pillow for "Enfeoffing feudal lords" (Jin Dynasty)(Collected by the Cizhou Kiln Art Museum in Anshi, Hebei Province)	Dragon-patterned plum vase (Jin Dynasty) (Collected by the Palace Museum)
			
Fish-patterned plat e(Yuan Dynasty) (Collection by the British Museum)	**Peony pattern jar (Song Dynasty) (Collected by the Palace Museum)**	**Figure pattern jar (Yuan Dynasty) (Collection of the Metropolitan Museum)**	**Fish-patterned bowl (Jin Dynasty) (Collected by the Palace Museum)**
			

Note: Permission for reproduction of all images has been obtained from the respective institutions.

Theoretical foundations

Situated cognition theory

Situated cognition theory posits that knowledge acquisition is an activity deeply embedded in sociocultural environments (Goodwin, 2000), rather than a simple information transmission process. This theory connects with cultural heritage transmission through three core points:

1. Authentic contexts: Learning in real or simulated environments is more effective thrning (Lave and Wenger, 1991; Giamellaro, 2017; Li et al., 2023; Troussas et al., 2024);
2. Social interaction: Learning is a social participation process that promotes understanding through interaction with others (Gasson and Waters, 2018; Nazlidou et al., 2024; Cosentino et al., 2025);
3. Practice participation: "Learning by doing" creates a deeper understanding than passive information reception (Danielsson and Linder, 2009).

Based on situated cognition theory, recent research demonstrates that experiential learning has significant advantages in cultural heritage transmission. Rather (2020)'s research proves that compared to traditional "knowledge-based" approaches, "experiential" participation increases cultural memory retention by 47%. Yu et al. (2024) developed the MakeBronze system that enhances children's understanding of bronze culture through hands-on experience, increasing participation willingness by 68%.

Co-creation theory

Co-creation theory originates from the service design field, emphasizing user participation in value creation processes (Ercsey, 2017). Applied to cultural heritage, co-creation presents three key characteristics:

1. Multi-party participation: Cultural transmission is no longer an exclusive activity of experts but a collaborative process among multiple parties (Paschen et al., 2020);
2. Value co-construction: Participants jointly construct understanding of culture through interaction processes (Ji, 2023);
3. Innovation integration: Creating new cultural expressions by fusing traditional and modern elements (Sipiran, 2024; Waidyaratne et al., 2024).

Recent research demonstrates that co-creation paradigms have unique value in cultural heritage protection.

Stefanidi et al. (2022) developed a virtual traditional craft platform that enhanced user understanding of traditional crafts through participatory design. Lee et al. (2019) promoted user exploration of cultural data through visual interactive systems, strengthening cultural identity.

Based on co-creation theory, and considering the integration of AI systems in contemporary digital heritage contexts, this research adopts the following core terminology:

"AI Collaborative Creation" refers to the process in which human users and AI systems jointly complete the creation of cultural content through multimodal interaction (textual descriptions, images, parameter adjustments), iterative refinement, and value co-construction. This term is distinct from "AI-assisted generation,"which emphasizes unidirectional content generation by AI, whereas AI collaborative creation emphasizes bidirectional interaction, mutual contribution, and shared creative agency between human and AI.This definition operationalizes co-creation theory in AI-enabled cultural heritage contexts, forming the theoretical foundation for our "Creation-as-Transmission" paradigm.

In summary, existing research has validated the respective values of situated cognition and co-creation theories, but lacks an integrative design framework that systematically combines both theories and is specifically designed to guide cultural heritage "learning and transmission through creation."

AIGC technology and AI collaborative creation system applications in cultural heritage protection

Artificial intelligence-generated content (AIGC) technology demonstrates two primary application directions in the field of cultural

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

heritage: cultural content generation, which emphasizes technical implementation, and interactive co-creation systems, which prioritize user engagement. This section provides a systematic comparison of the characteristics and limitations of these two application paradigms to identify research gaps directly relevant to this study.

Cultural content generation applications

Existing research primarily employs AIGC as a professional tool for the generation, restoration, and reconstruction of cultural content. These applications have achieved significant progress in technical implementation. For example, Yang (2024) combined Transformer and U-Net for Dunhuang mural restoration, achieving intelligent filling of damaged areas; Sun et al. (2024) applied Stable Diffusion to generate Tujia brocade patterns, providing creative inspiration for designers; Zeng (2024) utilized Midjourney to reconstruct 3D models of architectural heritage from historical photographs. Additionally, Liu (2020) developed MoCapGPT to capture traditional craft movements, while Smirnova et al. (2024) developed ancient manuscript text recognition systems. These technologies have expanded the application boundaries of AIGC in the cultural heritage protection field.

To systematically evaluate the characteristics and limitations of these applications, Table 2 provides a comparative analysis from dimensions including user participation, cultural fidelity assurance, and application objectives.

Comparative analysis reveals that while existing systems demonstrate superior technical performance, they exhibit significant deficiencies in cultural transmission. First, most systems position AIGC as auxiliary tools for professionals, where the general public can only passively receive generated content, thereby constraining the widespread dissemination of cultural heritage. Second, research predominantly focuses on optimizing generation quality (e.g., FID, PSNR metrics) while neglecting users' cultural cognition enhancement during interaction, lacking systematic evaluation of transmission effectiveness (Li et al., 2021). Furthermore, although systems are trained on cultural data, they lack effective constraints of cultural rules (Sun et al., 2024). These limitations reflect a prevalent disconnect between "technology-oriented" approaches and "transmission needs" in current systems. In response, this research proposes a "user-centered" participatory paradigm that systematically addresses issues of insufficient user engagement and weak educational functionality through platform-based tools, cognition-oriented evaluation, and rule-based generation.

AI collaborative creation system research

Compared to the unidirectional technical support of cultural content generation applications, AI collaborative creation systems emphasize bidirectional interaction between users and intelligent systems, providing participatory pathways for cultural transmission. In recent years, such systems have demonstrated potential across multiple domains, yet their application in cultural heritage contexts still faces critical challenges. Table 3 systematically compares key characteristics of representative AI collaborative creation systems in cultural heritage applications, revealing the strengths and limitations of existing systems.

The comparison in Table 3 reveals three core limitations of existing AI collaborative creation systems: (1) insufficient cultural adaptability: most systems (SmartPaint, ID.8) are not designed for specific cultural heritage; (2) polarized participation thresholds: professional systems (Reframer, DGBID) offer powerful functionality but restrict public participation; while MakeBronze (Yu et al., 2024), despite lowering the usage threshold through physical interaction, lacks depth in cultural knowledge transmission; (3) imbalance between cultural fidelity and innovation: difficulty in achieving equilibrium between cultural constraints and creative freedom. In contrast, this research overcomes the cultural adaptation deficiencies

Table 2. Comparison of this study with existing AI cultural generation systems for intangible cultural heritage

Research	Application scenarios	Core technologies	User participation methods	Cultural fidelity guarantee	Main limitations	Differences of this study
Sun et al. (2024)	Patterns of Tujia brocade	Stable diffusion	Limited (parameter adjustment)	Based on traditional pattern datasets	Lack of cultural rule constraints for verification	Embedded composition rules Real-time cultural feedback
Yang (2024)	Restoration of Dunhuang Murals	Transformer + U-Net	No (expert operation)	Training based on historical images	Restoration-oriented, noncreative participation	User-led creation Learning orientation
Zeng (2024)	3D reconstruction of architectural heritage	Midjourney + 3Dmodeling	None (operated by technicians)	Rely on historical photos	Technology-driven, not an educational goal	Equal emphasis is placed on education and inheritance Public participation
Liu (2020)	Traditional craft motion capture	MoCapGPT	No (recording tool)	Real motion capture	Recording tool, noninteractive system	Interactive creation experience
Smirnova et al. (2024)	Ancient book character recognition	OCR + Machine Learning	None (automated processing)	Based on language model	Identification tasks, noncreative participation	Creative expression Improvement of cultural cognition

Note:
- User participation method: None = Pure technical tools; Limited = Parameter adjustment; "Deep participation (in this studyUser participation method: None = Pure technical tools; Limited = Parameter adjustment; "Deep participation (in this study).
- Cultural fidelity guarantee: The main mechanisms by which each study maintains cultural characteristics.
- Main limitations: Insufficiency from the perspectives of user participation and cultural inheritance.
- Differences in this study: Highlighting the unique value of AIMagCreations.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

Table 3. Comparison of this study with existing AI collaborative creation systems

System name	Interaction method	Technical features	User group	Cultural fidelity	Creative freedom	Learning curve	Practical value
SmartPaint (Sun et al., 2019)	Stroke-level interaction	GAN generation	Designers	Low	High	Steep	Medium
MakeBronze (Yu et al., 2024)	Physical interaction	AR/VR	Children	Medium	Low	Gentle	High
ID. 8 (Antony and Huang, 2024)	Text-driven	Multimodal generation	Generaleral users	Low	Medium	Medium	Medium
DGBID (Deng et al., 2023)	Parameter adjustment	Bionic algorithms	Professional designers	Medium	High	Steep	High
KREA (Pei et al., 2024)	Eye-tracking control	Style transfer	Artists	Medium	Medium	Medium	Medium
Reframer (Lawton et al., 2023)	Stroke-level interaction	Collaborative interactive context-aware design agent (CICADA)	Professional creatives	Medium	High	Steep	Medium
Our Research	Multimodal interaction	SD-XI + LoRA	General users	High	Medium	Gentle	High

Note:
- **Culture-specific constraints:** Indicates whether the system incorporates built-in style/rule constraint mechanisms tailored to specific cultural heritage.
- **User threshold:** Determined based on target user groups and required prior knowledge (high = professionals, medium = experienced users, low = general public).
- Cultural fidelity mechanism: Technical methods extracted from original literature for preserving cultural characteristics.
- **Data source:** Method descriptions and experimental settings from the original papers of respective systems.

of general models through LoRA-based lightweight fine-tuning, maintains cultural depth while lowering participation thresholds through multi-level guided design, and provides a novel solution for AI collaborative creation systems in cultural heritage applications by establishing a dynamic balancing mechanism between cultural constraints and innovative expression, alongside systematic evaluation of cultural transmission efficacy.

HCI research in cultural heritage interaction design

Recent HCI research in cultural heritage interaction design shows four main trends: (1) immersive experience design, such as Southworth et al. (2024)'s VR cultural heritage experience system that enhances user engagement through spatial narrative; (2) collaborative cultural learning, such as Kim and Lee (2023)'s social learning framework promoting group cultural participation; (3) multimodal interaction, such as Hammady et al. (2019)'s mixed reality museum guide system integrating visual, auditory, and tactile feedback; (4) culturally adaptive interfaces, such as Martinez-Villareal et al. (2023)'s cross–cultural interaction design principles that adjust interaction modes based on cultural backgrounds.

Despite demonstrating the diversity of HCI applications in cultural heritage, three key gaps remain: First, most systems treat cultural elements as content carriers rather than core organizational principles of interaction design; Second, despite emphasizing participation, existing research rarely explores the unique value of creative activities as cognitive mediators; Finally, there is a lack of research frameworks that quantitatively assess cultural fidelity as a key system performance indicator. These gaps constitute the starting point of this research.

Research gaps and positioning of this study

Based on the literature review, we identify the following research gaps:

1. Theoretical integration gap: Existing research lacks integrative frameworks that organically combine situated cognition theory, co-creation theory, and cultural heritage transmission;
2. Design methodology gap: Lack of systematic interaction design methods specifically for cultural heritage transmission, particularly design principles balancing cultural fidelity and creative freedom;
3. Technical adaptation gap: General AIGC models lack a deep understanding and constraint capabilities for specific cultural heritage style characteristics;
4. Evaluation framework gap: Lack of standardized methods for systematically evaluating cultural transmission efficacy, making it difficult to compare the effectiveness of different interaction designs.

These interconnected gaps-theoretical integration, design methodology, technical adaptation, and evaluation framework-cannot be addressed through literature analysis alone. Therefore, we conducted empirical user research (Section "User Research and Design Framework Development") to identify practical challenges in Cizhou kiln cultural transmission, and synthesize these findings with theoretical insights to develop our conceptual framework (Section ""Creation-as-Transmission" Conceptual Framework").

User research and design framework development

Research methods

To quantitatively evaluate the current transmission status of Cizhou kiln culture among contemporary youth, we conducted multi-method research using stratified random sampling to recruit participants.

Participants: Total sample size n = 481, aged 18–45 years (M = 26.66), including 220 males (45.7%) and 261 females (54.3%), covering all 31 provinces, municipalities, and autonomous regions in China.

This research ensures sample representativeness and diversity through a multi-channel recruitment strategy combining online and offline approaches. Meanwhile, to enhance participation rates and data quality, this research implemented a combination of economic and non-economic incentives. Detailed recruitment channels, stratified sampling design, inclusion and exclusion

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

criteria, incentive measures, and experimental procedures are provided in Supplementary Appendix A.

Receipt collection period: October–November 2024.

Data collection: Combining quantitative and qualitative methods, including:

1. Structured questionnaire: Evaluating cultural cognitive levels, transmission channels, and participation motivations;
2. Semi-structured interviews (n = 20): In-depth exploration of participation barriers and interest points;
3. Observational records: Recording user interactions with existing cultural display formats.

Analysis methods: Using mixed analysis methods, including:

1. Descriptive statistics: Frequency analysis, percentage statistics;
2. Inferential statistics: Independent samples t-tests, analysis of variance;
3. Thematic analysis: Open coding of qualitative data to extract key themes.

Research findings

Basic cognitive status

The survey revealed that 43% (n = 208) of respondents had never been exposed to Cizhou kiln culture, with cognition showing significant stratification:

Historical background cognition rate: 86.3%.
Cultural value cognition rate: 59.6%.
Type characteristic cognition rate: 55.8%.
Representative work cognition rate: 48.1%.

These results indicate that even among those with basic cognition, a deep understanding of Cizhou kiln culture remains limited.

Transmission channel effectiveness

Table 4 data shows that although video resources are the main transmission channel (76.92%), retention rates are generally below 60%. Notably, interactive participation, despite having the lowest coverage (8.42%), achieved the highest user retention rate (82.60%), indicating that interactive experience has significant effects on maintaining user interest.

Participation barrier analysis

Through semi-structured interviews and open-ended responses, we identified three main types of participation barriers:

1. Technical barriers: 78% of respondents mentioned that Cizhou kiln creation "requires professional training," believing they lack necessary skills;
2. Cognitive distance: 63% of respondents felt traditional culture is "distant from modern life," struggling to find personal connections;
3. Display formats: 57% of respondents considered existing display methods "too static," lacking attractiveness and interactivity.

Design insights and principles

Based on user research findings and literature analysis, we extracted three core design principles (Table 5) as theoretical foundations for system development:

1. Progressive engagement principle: Design should lower initial participation barriers through multi-level structures while providing users with progressive deepening paths. This principle responds to Norman, D. (2013)'s "design visibility" concept but further emphasizes adaptive guidance in cultural learning contexts. Specific implementations include: (a) decomposing complex cultural creation into manageable steps; (b) providing contextualized presentation of cultural knowledge; (c) designing progressive disclosure interface elements.
2. Cultural embedding principle: Cultural knowledge should be integrated into interaction processes as design elements rather than content additions. This principle stems from situated cognition theory (Goodwin, 2000) but extends specifically for cultural transmission scenarios. Specific implementations include: (a) transforming cultural rules into interaction constraints; (b) implicitly conveying cultural values through design choices;

Table 4. Usage and retention rates of different transmission channels

Transmission channel	Usage rate (n = 273) (%)	Number	Initial interest	Continued participation	Retention rate (%)
Video resources	76.92	210	210	110	52.38
Print materials	50.18	137	137	52	37.96
Exhibitions	34.43	94	94	41	43.62
Lectures	26.74	73	73	32	43.84
Interactive participation	8.42	23	23	23	82.60

Table 5. Table corresponding to user challenges and design principles

User challenge	Framework mechanism	Design principles	Expected results
High participation barrier (78% perception technology difficulty)	Multi-level guidance design	Progressive participation	Reduce cognitive load through phased interaction
Cognitive gap (43% have never been exposed to Cizhou Kiln)	Cultural embedding architecture	Achieve cultural fidelity through constraints	Implicit learning that embeds knowledge through interaction
Low inheritance efficiency (>60% retention rate of traditional methods)	Socialized inheritance cycle	Creation Fidelity balance	Enhance engagement through active participation

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

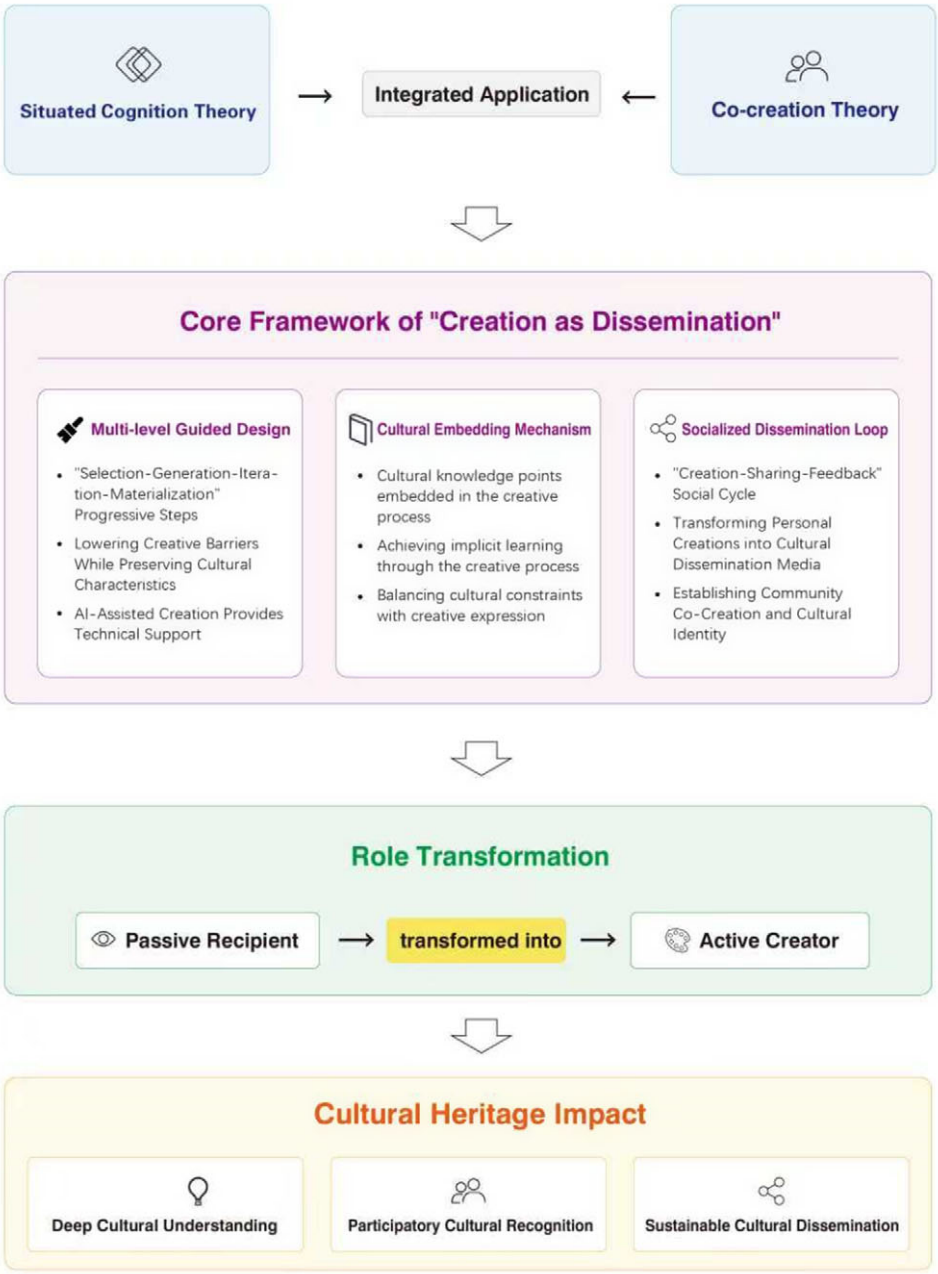


Figure 1. Creation-as-transmission conceptual framework. *Note*: This framework integrates situated cognition theory and co-creation theory through three core mechanisms: multi-level guided design, cultural embedding architecture, and social transmission loop. Arrows indicate causal pathways from theoretical foundations (bottom) to design mechanisms (middle) to user role transformation (top). Detailed explanations of each component are provided in Section "Creation-as-Transmission" Conceptual Framework".

(c) creating immediate associations between cultural knowledge and user behavior.

3. Creation-fidelity balance principle: Systems must seek balance between user creative freedom and cultural characteristic fidelity. This principle echoes discussions of "controlled freedom" in HCI (Polidoro et al., 2024). Specific implementations include: (a) designing adjustable cultural constraint mechanisms; (b) providing culture-grammar-based creation assistance; (c) constructing cultural feedback loops.

"Creation-as-transmission" conceptual framework

Synthesizing the research gaps identified in Section "Research Gaps and Positioning of This Study" and the design insights derived from user study (Section "Design Insights and Principles"), we propose the "Creation-as-Transmission" conceptual framework (Figure 1) as a systematic solution to the identified challenges.

Framework overview and theoretical positioning

The framework integrates situated cognition theory and co-creation theory to address the core challenge of cultural heritage transmission: transforming users from passive recipients to active co-creators.

Synergistic integration: Rather than treating these theories separately, our framework achieves synergy by positioning co-creation as the method and situated cognition as the mechanism. Creative participation (co-creation) generates authentic contexts (situated cognition) where cultural learning naturally occurs. This integration addresses the research gap of systematic theoretical application

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

in cultural heritage interaction design (Section "Research Gaps and Positioning of This Study").

Three-layer architecture (Figure 1): The framework comprises (1) theoretical foundation: situated learning principles + co-creation paradigm; (2) core mechanisms: translating abstract principles into concrete design mechanisms; (3) outcomes: user role transformation.

Core design mechanisms

The framework operationalizes the design principles (Section "Design Insights and Principle") through three interconnected mechanisms:

1. Mechanism 1—Multi-level guided design: Implements a four-stage workflow-selection, generation, iteration, and materialization-that reduces cognitive load while maintaining cultural depth by decomposing complex cultural creation into manageable steps (Norman, 2013).
2. Mechanism 2—Cultural Embedding Architecture: Integrates cultural rules as interaction constraints and embeds knowledge cards within the workflow, enabling implicit learning through practice by transforming cultural knowledge from explicit instruction to embodied experience (Goodwin, 2000).
3. Mechanism 3—Social Transmission Loop: Establishes a creation-sharing-feedback cycle with cultural validation, transforming individual creation into socialized cultural dissemination through social participation that enhances cultural understanding (Lave and Wenger, 1991).

Framework validation: mapping challenges to solutions

To verify the applicability of the framework, we map three user research challenges to specific framework components:

AIMagCreations system design and implementation

System architecture overview

Building upon the "Creation-as-Transmission" framework (Section "Creation-as-Transmission" Conceptual Framewok", Figure 1), we developed the AIMagCreations system to operationalize the three core mechanisms. Figure 2 illustrates the system architecture, which integrates three main levels: data layer, algorithm layer, and interaction layer, forming a complete technical implementation framework.

Data layer: Provides 6,000 annotated Cizhou kiln pattern samples with multi-dimensional cultural attributes (vessel types, themes, compositional rules), supporting cultural knowledge embedding (Section "Data Layer Design").

Algorithm layer: Implements VAE + LoRA-based generation pipeline (compression ratio 1/64, LoRA rank r=64) with cultural constraint validation every 10–20 steps, achieving 89.4% cultural fidelity and FID = 21.7 (Section "Algorithm Layer Implementation").

Interaction layer: Realizes a four-stage co-creative workflow (selection → generation → iteration → materialization) with multimodal inputs and social transmission functions.

Data layer design

The system first constructs a complete Cizhou kiln cultural data asset repository, covering multi-dimensional characteristics including morphological classification, pattern semantics, composition rules, and cultural attributes.

Data collection

Using multi-source fusion strategies to ensure data comprehensiveness and authority:

1. Field collection: Collaborating with Cizhou kiln intangible heritage inheritors for field research, using high- precision 3D scanners (0.1 mm accuracy) to obtain original artifact data;
2. Literature digitization: Integrating authoritative literature such as "Cizhou Kiln Porcelain Pillows" line drawings, high-resolution photography (600dpi) of museum-housed Cizhou kiln artifacts;
3. Network resource integration: Crawling public images from cultural heritage databases, with expert team review and screening.

Through these methods, 13 TB of raw data were obtained. Using the perceptual image hashing algorithm (ImageHash) to remove duplicate images (similarity > 95%), manually eliminating low-resolution (< 1024 × 1024), damaged, or obviously restored samples, finally retaining 6,000 high-quality pattern images.

Data annotation and structuring

Multi-dimensional annotation of pattern data to construct a structured database:

The annotation process adopts expert review mechanisms to ensure annotation quality and cultural accuracy. The final database contains 6,000 high-quality samples covering the main Cizhou kiln vessel types and pattern categories. Cizhou Kiln data annotation statistics are presented in Table 6.

Algorithm layer implementation

This research proposes a generative AI-based intelligent generation algorithm framework for Cizhou kiln decorative patterns. The system uses Stable Diffusion-XL as the base model, integrating core algorithm modules including image perceptual compression, composition template guidance, and style transfer.

Multi-scale image perceptual compression

To optimize computational efficiency while maintaining pattern details, we designed a multi-scale image perceptual compression algorithm based on variational autoencoder (VAE). The algorithm progressively compresses images through multiple scales, reducing computational complexity from original space to 1/64 when downsampling factor f=8.

Key technical specifications:

(1) Architecture: VAE encoder-decoder with multi-scale processing
(2) Loss function: MSE reconstruction loss + KL divergence regularization
(3) Performance: Single inference time 12.3 ± 2.1 s (NVIDIA RTX 3090 GPU)

Table 6. Cizhou Kiln data annotation statistics

Annotation dimension	Sample scale	Coverage
Morphological classification	10 vessel types	Plates, bowls, jars, etc.
Pattern semantics	6 major themes	Blessing, academic success, etc.
Composition rules	2 basic compositions	Symmetrical, asymmetrical
Cultural attributes	3 dynasties	Song, Jin, Yuan

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

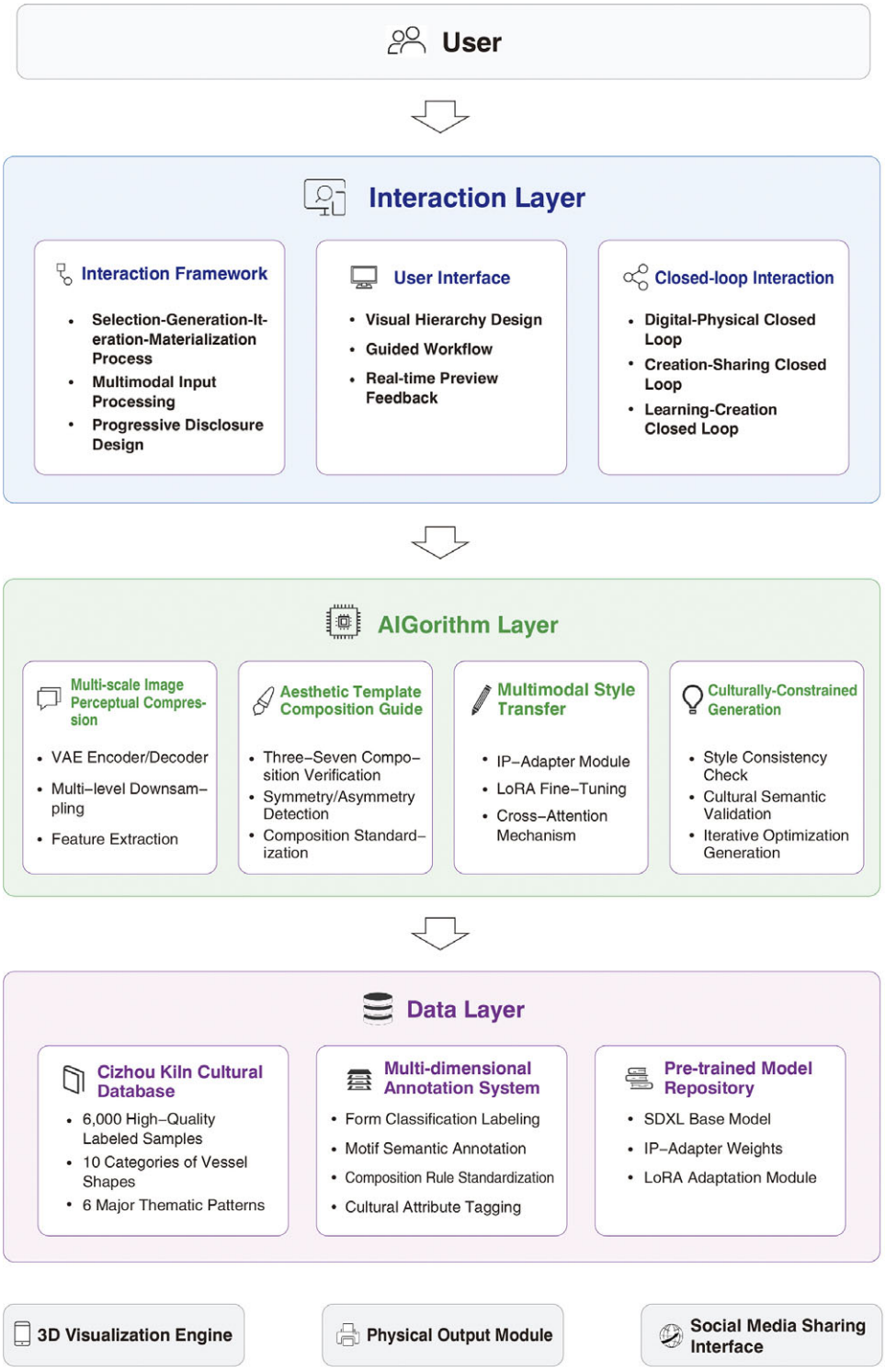


Figure 2. AIMagCreations system architecture diagram. *Note*: The system architecture operationalizes the "Creation-as-Transmission" framework (Figure 1) through three integrated layers. The Data layer provides a structured Cizhou kiln cultural repository with multi-dimensional annotations (Section "Data Layer Design"). The algorithm layer implements the VAE + LoRA-based multimodal generation framework with cultural constraints, ensuring style fidelity (FID = 21.7) and cultural authenticity (89.4% fidelity) (Section "Algorithm Layer Implementation"). The interaction layer realizes the four-stage co-creative workflow (selection*I* →generation*I* →iteration*I* →materialization), translating the multi-level guided design mechanism into concrete user interactions (Section "Interaction Layer Design"). Arrows indicate data flow and functional dependencies between layers.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

(4) Compression ratio: 1/64 while preserving cultural pattern details

Detailed algorithm implementation is provided in Supplementary Appendix E1.

Aesthetic template-based composition guidance

To maintain typical Cizhou kiln composition characteristics (such as "three-seven composition"), we designed a template-based composition guidance mechanism that constrains the generation process through cultural rule templates.

Key design principles:

(1) Template representation: Binary masks encoding composition rules ($M[i,j] \in \{0,1\}$).
(2) Adaptive denoising: Adjustable denoise strength α controls creative freedom
(3) Cultural validation: Automatic verification of "three-seven ratio" and symmetry rules.

This mechanism ensures generated content follows traditional composition rules while preserving creative space. Implementation details are provided in Supplementary Appendix E2.

Multimodal style transfer

Combining IP-Adapter and LoRA technologies for efficient style transfer, addressing the lack of cultural specificity in general models: Core technical components:

1. IP-Adapter module: Achieves multimodal feature fusion through decoupled cross-attention mechanisms, enabling the model to simultaneously process text descriptions and reference images. Uses CLIP ViT–L/14 as image encoder, cross-attention weights set to 0.8, with an additive attention mechanism for feature fusion;
2. LoRA fine-tuning strategy: Uses low–rank decomposition method (r = 64) for efficient adaptation of pre-trained models, significantly reducing computational costs while maintaining style consistency. Specific configurations include learning rate 2e–4 (cosine scheduling), 30,000 training steps, batch size 16, using AdamW optimizer ($\beta1$ = 0.9, $\beta2$ = 0.999, weight_decay = 0.01).

Technical implementation configuration: Based on Stable Diffusion–XL v1.0 (resolution 1024 × 1024) as base model, trained on 4 × NVIDIA A100 GPU (80GB), 128GB RAM hardware environment, using CUDA 11.8 and PyTorch 2.0.1. These parameter configurations achieve optimal style transfer effects while maintaining computational efficiency, particularly LoRAIs low-rank constraint (r = 64) significantly reducing computational complexity while maintaining Cizhou kiln characteristics.

Performance evaluation results: Quantitative assessment shows this method reduces training time from traditional full–parameter fine–tuning's 48.7 hours to 12.3 hours (74.7% reduction) while achieving precise cultural characteristic preservation. All code and pre–trained models will be publicly released after publication to ensure research reproducibility.

Cultural constraint generation algorithm

To ensure generated content conforms to Cizhou kiln cultural characteristics, we designed a multi-condition constrained generation process that integrates cultural rules throughout the generation pipeline:

Constraint mechanisms:

1. Composition constraints: Validation of "three-seven ratio" (subject-to-background $\approx 3:7$).
2. Style constraints: Verification of black-white contrast characteristics.
3. Semantic constraints: Cultural meaning validation of pattern elements the generation process employs iterative refinement with periodic cultural validation (every check_interval steps) to ensure both technical quality and cultural authenticity. The complete algorithm workflow is detailed in Supplementary Appendix E3.

Algorithm performance evaluation

Algorithm performance was evaluated through comparative experiments, with results shown in Table 7.

Experimental results show that by introducing cultural feature constraint mechanisms, the system maintains traditional Cizhou kiln composition rules (92.7% accuracy) and cultural feature fidelity (89.4% accuracy) while maintaining fast generation speed (average 15.0 seconds).

Interaction layer design

Interaction framework design

The core innovation of AIMagCreations lies in the "Creation-as-Transmission" interaction paradigm, achieved through a four-stage AI collaborative creation framework: selection, generation, iteration, and materialization (Figure 3).

1. Selection phase: Users explore cultural elements through three entry points:
   - Theme selection: Choose from six traditional themes (prosperity, academic achievement, etc.);
   - Form selection: Browse 10 classic vessel types, learning historical backgrounds;
   - Pattern elements: Decorative element library organized by symbolic meaning.
2. Generation phase: Transform multimodal inputs into Cizhou kiln style designs:
   - Multimodal input: Support text descriptions, uploaded images, or both;
   - Cultural constraints: Automatically apply Cizhou kiln composition rules;
   - Visual feedback: Real–time visualization maintains user engagement.

Table 7. Pattern generation performance evaluation

Evaluation metric	Baseline model	Our method	Improvement
FID	58.3	21.7	62.8%
PSNR (dB)	32.1	38.6	20.2%
Composition accuracy	75.3%	92.7%	23.1%
Cultural feature fidelity	68.5%	89.4%	30.5%
Generation time (s)	45.2	15.0	66.8%

Notes: (1) FID (Fréchet Inception Distance): Lower is better, <30 indicates excellent performance. (2) PSNR (Peak Signal-toNoise Ratio): Unit is dB, $>35dB$ indicates high-quality reconstruction. (3) Compositional accuracy: The conformity to "threeseven composition" rules independently annotated by 3 experts, Fleiss' Kappa = 0.83. (4) Cultural feature fidelity: Weighted average of three sub-dimensions including black-white contrast, symmetry, and decorative semantics. (5) Baseline model: Stable Diffusion-XL v1.0 original model (not optimized for Cizhou kiln).

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

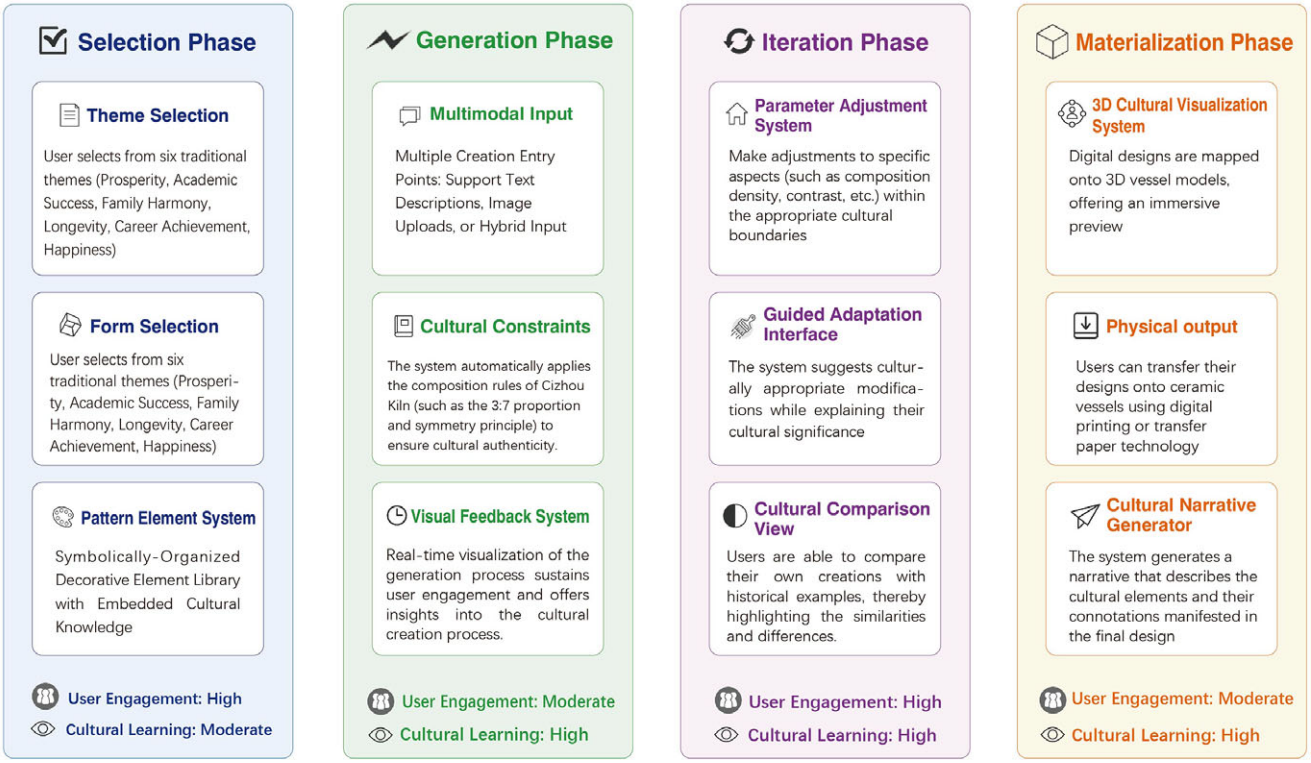


Figure 3. "Selection-generation-iteration-materialization" AI collaborative creation framework.

3. Iteration phase: Support guided refinement:
   - Parameter adjustment: Modify specific aspects within culturally appropriate ranges;
   - Guided adaptation: System suggests culturally appropriate modifications;
   - Comparative view: Compare with historical examples.
4. Materialization phase: Transform digital designs into physical objects:
   - 3D visualization: Map to 3D vessel models;
   - Physical output: Transfer to ceramic vessels through digital printing or transfer paper techniques;
   - Creation story: Generate narratives describing cultural elements in the design.

User interface design

The AIMagCreations interface implements the four-stage framework through progressive disclosure methods,maintaining creative freedom while guiding users (Figure 4). The interface design particularly emphasizes balance between cognitive load management and cultural guidance, adopting a three-layer information architecture: (1) interactive layer, providing direct operation interfaces; (2) cultural knowledge layer, providing cultural backgrounds through contextual prompts and embedded information cards; (3) creation guidance layer, providing timely creation suggestions. This multi-layer architectural design responds to Norman (2013)'s "knowledge in the world" design concept, embedding cultural knowledge not only in user cognition but also in interface elements.

Key interface design features include:

1. Progressive disclosure architecture with cognitive load management staged information architecture: Implements a four-phase workflow design following Nielson's progressive disclosure principles, with each interface optimized for specific cognitive tasks to minimize extraneous cognitive load (Sweller, 2011).

Hierarchical task decomposition: Employs modular interface segregation where complex cultural design tasks are decomposed into manageable sub-tasks, supporting users' working memory limitations.

Contextual scaffolding: Integrates just-in-time cultural knowledge delivery through embedded information cards and contextual tooltips, reducing split-attention effects.

2. Multimodal interaction design with cultural constraint integration

Input modality convergence: Implements parallel input channels (textual, visual, template-based) within a unified interaction space, supporting diverse user preferences and expertise levels real-time visual feedback loop: Employs immediate preview mechanisms in the central workspace to maintain user agency and reduce gulf of evaluation (Norman, 2013).

Culturally informed affordances: Integrates traditional aesthetic constraints as interface affordances rather than restrictions, making cultural rules discoverable through interaction design.

3. Distributed cognition framework for cultural knowledge

External memory support: Implements Norman's "knowledge in the world" paradigm by embedding cultural semantics directly into interface elements, reducing reliance on user's prior cultural knowledge.

Semantic layering architecture: Employs three-tier information architecture: (1) direct manipulation layer, (2) cultural context layer, and (3) generative guidance layer, supporting both novice and expert users.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

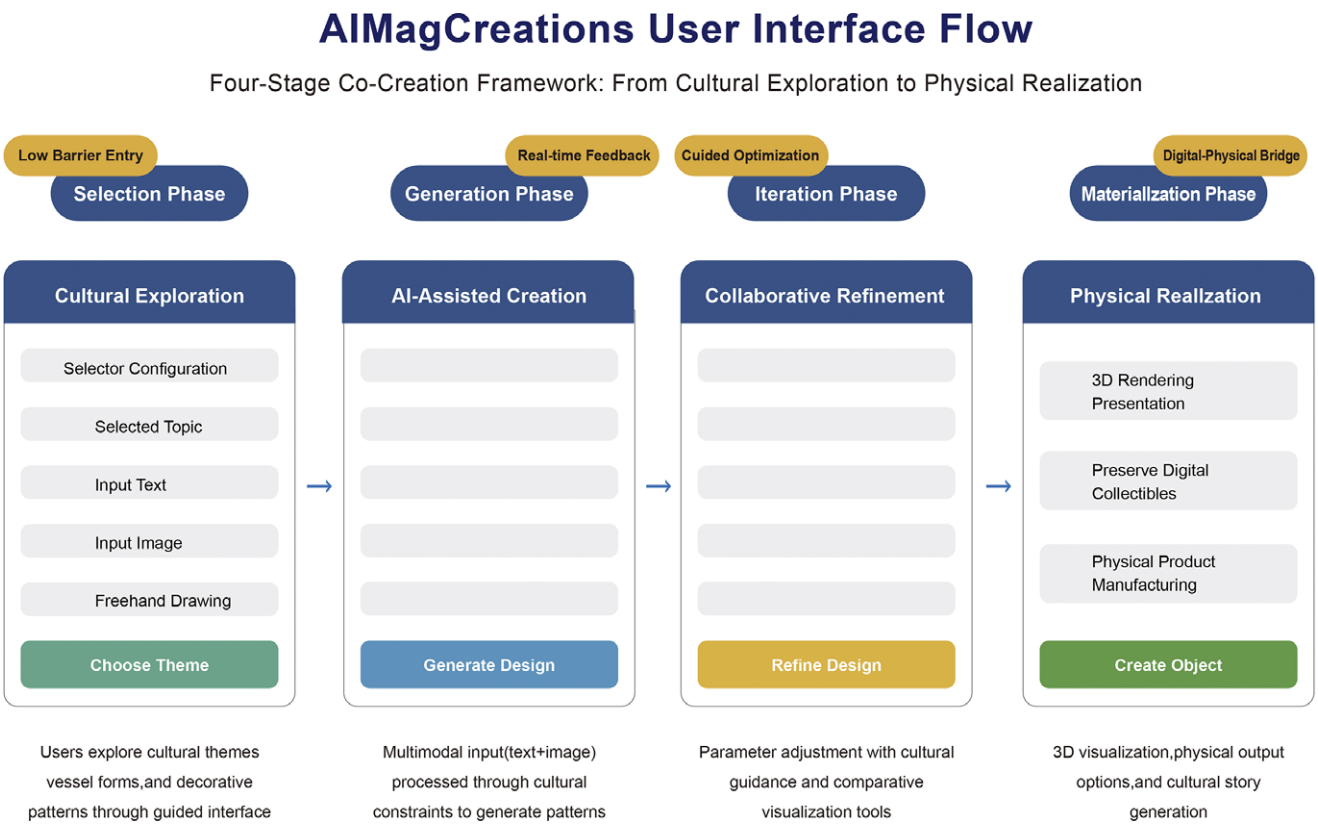


Figure 4. AIMagCreations user interface flow.

Cultural sensemaking support: Provides comparative visualization tools and historical exemplar references to support users' cultural pattern recognition and appropriation processes.

4. Adaptive automation with human agency preservation

Mixed-initiative design: Balances AI-driven cultural constraint application with user creative control through transparent system suggestions and override capabilities.

Granular control mechanisms: Implements progressive complexity options (primary patterns, auxiliary patterns, decorative elements) allowing users to engage at their preferred level of detail.

Explainable AI integration: Makes algorithmic cultural adaptations visible through interface feedback, supporting user understanding and trust calibration.

5. Social cognitive design for cultural transmission

Community-integrated workflow: Seamlessly integrates individual creation with social sharing mechanisms, supporting cultural knowledge propagation through peer learning.

Digital-physical bridging: Implements tangible interaction endpoints (QR codes, physical output options) that extend digital cultural creation into material culture practices.

Collaborative affordances: Provides social comparison tools and community integration points that support collective cultural meaning-making processes.

Design rationale integration: This interface design operationalizes Activity Theory principles by treating cultural ceramic design as a mediated activity system where traditional knowledge, digital tools, and user creativity form an integrated sociotechnical ensemble. The progressive disclosure approach specifically addresses the challenge of cultural HCI identified by Winschiers-Theophilus et al. (2022) - maintaining cultural authenticity while enabling creative appropriation through technology mediation.

Figure 5 complete digital–physical cultural transmission ecosystem demonstrating three interconnected loops. (a) AI collaborative creation mode enables cultural learning through theme selection, multimodal input processing, and intelligent pattern rendering. The ecosystem implements a complete digital-physical loop by seamlessly connecting virtual creation with physical manifestation, supporting both individual cultural learning and collective cultural preservation. This comprehensive approach transforms traditional unidirectional cultural transmission into a participatory, cyclical experience that reinforces cultural connections through both digital engagement and tangible ownership. (b) Physical generation mode materializes digital designs into tangible ceramic products through 3D preview, cultural information display, digital asset preservation via QR code, and physical production ordering.

Interaction loop construction

To maximize cultural transmission effects, three interconnected interaction loops were designed:

1. Digital-physical loop: Connecting virtual creation with physical objects through ceramic transfer technology;
2. Creation-sharing loop: Promoting social sharing of creation processes and final works;
3. Learning-creation loop: Embedding cultural knowledge in creation processes for implicit learning.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

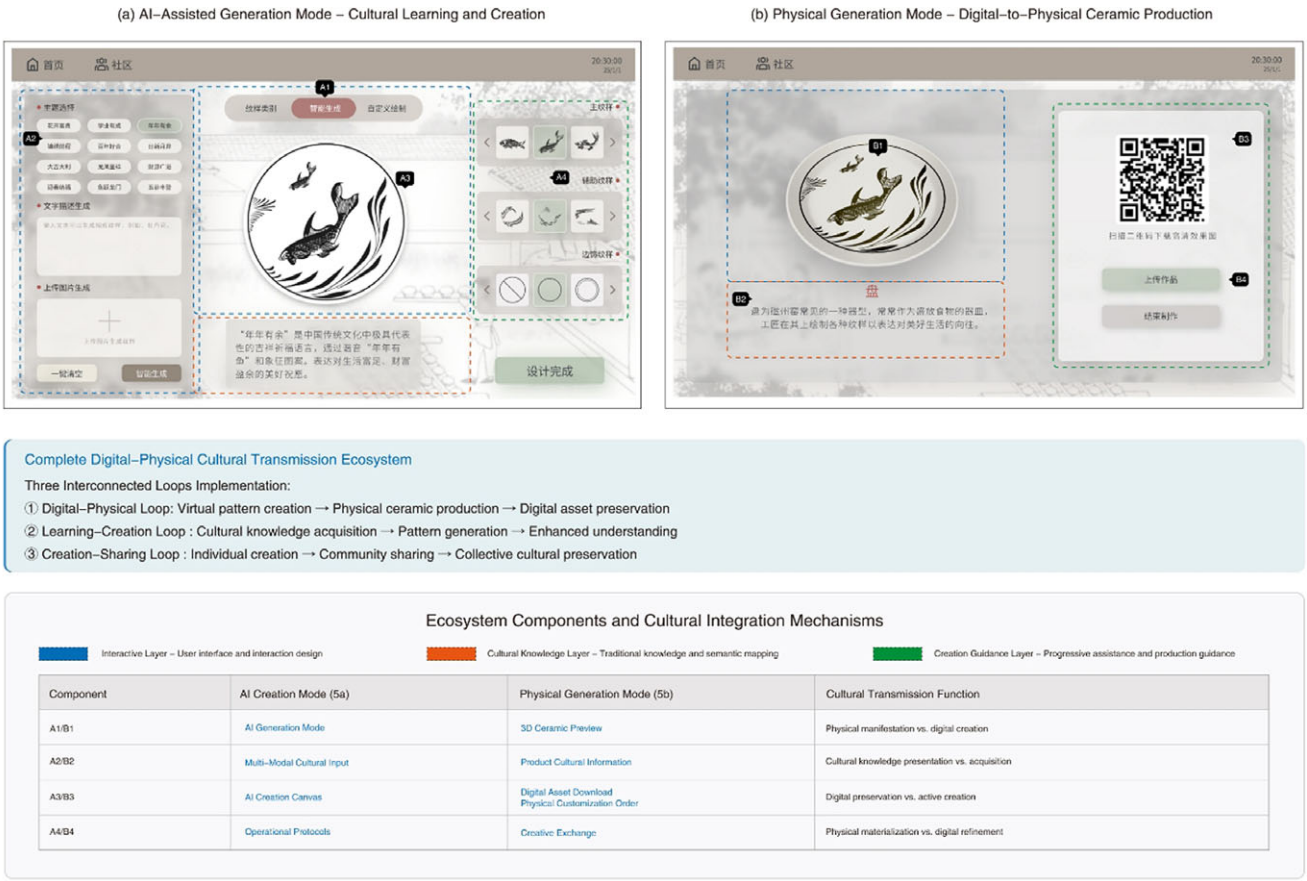


Figure 5. Complete digital-physical cultural transmission ecosystem.

These interconnected loops transform traditional unidirectional cultural transmission into cyclical participatory experiences, continuously reinforcing cultural connections.

System evaluation: multi-dimensional assessment

Evaluation framework

To comprehensively evaluate the effectiveness of the design framework, we developed a four-dimensional evaluation method (Table 8), validating system performance and effects from different perspectives.

The four studies form a progressive logic of "function validation – efficacy assessment – need insight" around different dimensions. Detailed participant information is in Supplementary Appendix C, with detailed experimental procedures in Supplementary Appendix D.

Statistical analysis methods

All statistical analyses were conducted using established procedures with appropriate assumption testing and effect size reporting.

Descriptive statistics: Central tendency and dispersion were summarized using means, standard deviations, and 95% confidence intervals. Normality was assessed via Shapiro–Wilk tests, and outliers were identified using boxplots (1.5× IQR criterion).

Between-group comparisons: Independent samples *t*-tests compared experimental and control groups after verifying assumptions

Table 8. Four-dimensional evaluation framework

Evaluation type	Evaluation method	Evaluation purpose	Participants
Expert evaluation	System Usability Scale (SUS) (AlGhannam et al., 2017) (n = 20) (Supplementary Appendix B1; Bastien, 2010)	Validate system usability	HCI, AI, and design domain experts
Technical performance evaluation	Ablation experiments (complete/w/o LoRA/full fine-tuning) (Supplementary Appendix B2)	Validate system technical performance	AI and computer vision experts (n = 20)
Cultural transmission efficacy evaluation	Mixed experiments (n = 100, traditional vs system groups)	Validate system effectiveness in cultural transmission	18–24 year-old youth users
Co-creation experience research	Mixed Active Creativity Support Index MICSI (Lawton et al., 2023) (n = 60) + in-depth interviews (Supplementary Appendix B)	Deep exploration of user needs	Multidisciplinary undergraduate students

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

(normality and homogeneity of variance via Levene's test). When assumptions were violated, Welch's *t*-test or non-parametric Mann–Whitney *U* tests were employed as appropriate.

Repeated measures analysis: Within-subjects comparisons used repeated measures ANOVA with Greenhouse–Geisser correction for sphericity violations (assessed via Mauchly's test). Post-hoc pairwise comparisons employed Bonferroni correction.

Multiple testing correction: To control family-wise error rates, Bonferroni correction was applied systematically: α_adjusted $= \alpha$ / number of tests.

Effect sizes: Cohen's d quantified standardized mean differences (small: $d = 0.2$, medium: $d = 0.5$, large: $d = 0.8$). Pearson correlation coefficients (r) assessed association strength (weak: $|r| < 0.3$, moderate: $0.3 \leq |r| < 0.6$, strong: $|r| \geq 0.6$).

Regression analysis: Hierarchical multiple regression examined predictive relationships, with thorough diagnostic testing (linearity, independence, homoscedasticity, normality of residuals, multicollinearity via VIF). Standardized beta coefficients enabled cross-predictor comparisons.

Software and reproducibility: Analyses employed SPSS Statistics 27.0 and R version 4.2.1 (packages: ez, emmeans, car, lmtest, QuantPsyc). All random processes used specified seeds for reproducibility. Significance level was set at $\alpha = 0.05$ unless otherwise noted.

Table 9. SUS scores in each dimension

SUS item	Item content	Mean score	Standard deviation	95% CI	Score range
Q1	I would like to use this system	4.25	0.44	[4.04,4.46]	4–5
Q2	I think this system is complex	1.85	0.59	[1.58,2.12]	1–3
Q3	I think this system is easy to use	4.25	0.55	[3.99,4.51]	3–5
Q4	I need technical support to use this system	2.05	0.51	[1.81,2.29]	1–3
Q5	I found the system functions well integrated	3.85	0.49	[3.62,4.08]	3–5
Q6	I think the system is inconsistent	1.45	0.51	[121,1.69]	1–2
Q7	I think most people could learn to use this system quickly	4.25	0.64	[3.95,4.55]	3–5
Q8	I find this system cumbersome to use	1.75	0.44	[1.54,1.96]	1–2
Q9	I feel confident using this system	3.95	0.39	[3.77,4.13]	3–5
Q10	I need to learn many things before using this system	2.00	0.46	[1.79,2.21]	1–3

Notes: Each dimension scored on 5-point scale, overall SUS score on 100-point scale Q2, Q4, Q6, Q8, Q10 are reverse items, scores reverse-calculated Overall SUS Score: 78.63 ± 7.41 (range: 65–92.5, $n = 20$).

Expert usability evaluation

Experimental design

Twenty experts from interaction design, cultural heritage, and education fields (average age 40.35 ± 2.54 years, 50% female) participated in the evaluation. The experiment used standardized tasks and the System Usability Scale (SUS), with procedures including:

1. Pre-experiment preparation (10 minutes): Sign informed consent forms, complete background survey questionnaires, watch system introduction;
2. System experience (40 minutes): Complete guided tasks (10 minutes), free exploration (10 minutes), deep creation (20 minutes);
3. Evaluation feedback (30 minutes): Complete SUS questionnaire, participate in in-depth interviews.

Results analysis

Expert evaluation results (Table 9) show the system has good usability, specifically demonstrated in:

1. Quantitative evaluation results: System achieved average SUS score of 78.625 (SD = 7.411), significantly higher than "good" usability threshold of 70 points ($t(19) = 5.23$, $p < 0.001$, Cohen's $d = 1.17$);
2. Function completion: 88% of experts independently completed all tasks (M duration = 18.5 min, SD = 3.2), with key operation error rate below 5%;
3. Expert feedback themes: Through thematic analysis of interview data, three core feedback categories emerged:
    - Interface design advantages (mention rate 72%): Interaction flow follows intuition, visual guidance reduces learning costs;
    - Technical efficacy recognition (mention rate 65%): Real–time generation speed (< 15s) and high style transfer fidelity;
    - Improvement suggestions (frequency > 3 times): Add preset templates (5 people), optimize loading prompts (3 people).

Technical performance validation

Experimental design

To validate the critical role of the LoRA module in Cizhou kiln pattern style transfer, three control experiments were designed:

1. Complete model: Contains complete VAE encoder, IP-Adapter, and LoRA modules;
2. Without LoRA model: Removes LoRA module, retaining only base Stable Diffusion-XL model;
3. Full parameter fine-tuning model: Full parameter fine-tuning of UNet network using the same training dataset.

Table 10. Three model technical metrics comparison

Model type	FID↓	PSNR (dB)↑	Training time (h)↓	Memory usage (GB)↓
Complete model	21.7	38.6	12.3	5.3
w/o LoRA	58.3	32.1	10.1	5.1
Full fine-tuning	24.5	37.8	48.7	15.8

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

Table 11. User experience score comparison (mean ± SD)

Evaluation dimension	Complete model	Without LoRA	Full fine-tuning
Visual aesthetics	6.0 ± 0.8	3.9 ± 1.2	5.8 ± 0.9
Cultural element recognition	6.3 ± 0.7	3.7 ± 1.1	5.7 ± 0.8
Style consistency	6.2 ± 0.8	3.9 ± 1.2	5.8 ± 0.9
Detail fidelity	5.9 ± 0.7	3.7 ± 1.1	5.7 ± 0.8
Innovation level	5.8 ± 0.9	4.2 ± 1.0	5.5 ± 0.9
Overall satisfaction	6.1 ± 0.7	4.0 ± 1.1	5.8 ± 0.8

For each configuration, random sampling from 600 test images for evaluation, with 20 professional evaluators (average age 38.45 ± 3.21 years) conducting technical metrics and subjective experience assessments. Technical metrics results and subjective experience assessments results are presented in Table 10 and Table 11.

Assumption testing. Prior to between-group comparisons, Levene's test assessed homogeneity of variance across model configurations. For FID comparisons, the test revealed unequal variances between complete and w/o LoRA models ($F(1,1198) = 4.23$, $p = 0.040$), necessitating Welch's t-test correction. Similarly, PSNR comparisons showed variance heterogeneity ($F(1,1198) = 3.87$, $p = 0.049$). Comparisons with the Full Fine-tuning model met variance homogeneity assumptions (FID: $F(1,1198) = 0.89$, $p = 0.346$; PSNR: $F(1,1198) = 0.65$, $p = 0.420$), permitting standard t-tests. For subjective evaluations (Table 11), Levene's tests confirmed homogeneous variances across all six dimensions (all $F(2,57) < 2.34$, all $p > 0.105$), satisfying ANOVA prerequisites.

Key findings include:

The ablation study validates the critical role of the LoRA module from both technical performance and user perception perspectives:

(1) Technical performance improvements

1. Style fidelity: Removing the LoRA module caused FID to surge from 21.7 to 58.3 ($+168\%, t = 9.42$, $p < 0.001$), demonstrating LoRA's indispensable role in capturing Cizhou kiln-specific stylistic features;
2. Detail preservation: The complete model achieved significantly higher PSNR (38.6 dB) than the w/o LoRA model (32.1dB, $p < 0.001$), confirming that low-rank constraints ($r = 64$) effectively mitigate detail degradation during style transfer;
3. Computational efficiency: Compared to full fine-tuning, LoRA reduced training time by 74.7% (12.3h vs. 48.7 h) and memory footprint by 66.5% (5.3 GB vs. 15.8 GB, significantly enhancing system deployability while maintaining generation quality.

(2) User perception validation

1. Visual aesthetics: The complete model scored significantly higher (M=6.0 ± 0.8) than the w/o LoRA model (M=3.9 ± 1.2, $p < 0.001$), with no significant difference from full fine-tuning (M=5.8 ± 0.9, p=0.18), indicating LoRA achieves comparable visual quality to full-parameter approaches;
2. Cultural element recognition: The complete model received the highest rating (M=6.3 ± 0.7), significantly outperforming both the w/o LoRA model (M=3.7 ± 1.1, $p < 0.001$) and full fine-tuning (M=5.7 ± 0.8, $p < 0.01$), suggesting LoRA's cultural constraint mechanism surpasses generic training strategies;
3. Overall cultural fidelity: Across three dimensions-style consistency, detail preservation, and cultural recognition-the complete model averaged 6.13, exceeding the w/o LoRA model by 57.4% ($p < 0.001$) and full fine-tuning by 6.6% ($p < 0.05$).

(3) Metric-perception consistency

Objective technical metrics demonstrated strong alignment with subjective perceptual evaluations. Correlation analyses revealed that FID scores exhibited a strong negative correlation with visual aesthetics ($r = -0.82$, $p < 0.001$), while composition accuracy showed a strong positive correlation with cultural element recognition (r=0.79, $p < 0.001$). This metric-perception consistency demonstrates that: (a) the proposed cultural constraint mechanism not only optimizes computational objectives but also aligns closely with human aesthetic judgment and cultural cognition; (b) the LoRA module achieves optimal balance across efficiency, quality, and cultural fidelity through precise cultural feature modeling.

Cultural transmission efficacy evaluation

Experimental design

We conducted controlled experiments with 100 participants aged 18 – 24 years (average age 21.40 ± 2.35 years, balanced gender ratio), Participants were randomly assigned to experimental ($n = 50$) or control ($n = 50$) groups using computer-generated random number sequences. The experimental scenarios are shown in Figure 6 and Figure 7. Randomization employed a stratified block design to balance key variables including gender, age, and cultural background. A priori statistical power analysis (GPower 3.1.9.7) indicated that the sample size of $n = 100$ provides >95% power to detect anticipated effects ($d = 0.70$) at $\alpha = 0.05$. Detailed randomization procedures, power analysis, baseline equivalence verification, and sensitivity analyses are provided in Supplementary Appendix F. Comparing AIMagCreations with traditional learning methods (lectures and videos):and exploring the relationship between creative behaviors and cultural cognition.



Figure 6. Control group (explanation picture of Cizhou Kiln).

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

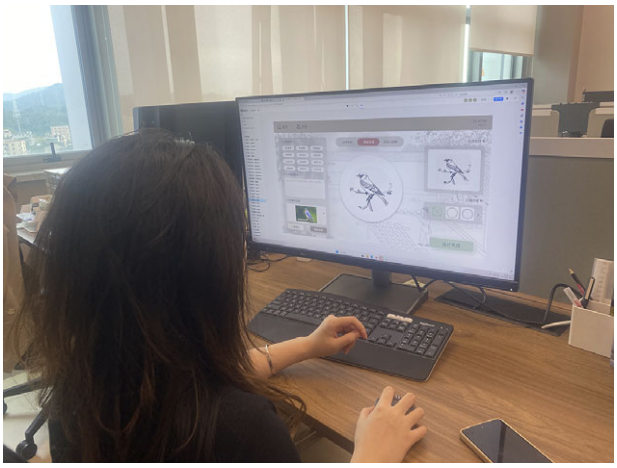


Figure 7. Experimental group (experimental process diagram).

1. Experimental design: Between-subjects design, random assignment to two groups:
   - Experimental group (n = 50): System function video tour (8 minutes), free creation experience (20 minutes), complete evaluation questionnaire (2 minutes);
   - Control group (n = 50): Cultural explanation video (8 minutes), themed lecture (20 minutes), complete evaluation questionnaire (2 minutes).
2. Measurement indicators:
   - Knowledge acquisition: Five standardized questions evaluating cognitive levels (Supplementary Appendix B3 Questionnaire 1);
   - Cultural interest: Four scales measuring learning enjoyment and other indicators (Supplementary Appendix B3 Questionnaire 2);
   - Behavioral indicators: Sharing behavior, exploration patterns, and subsequent information seeking.
   - Behavioral indicators: Creative engagement indicators, interaction pattern indicators, socialized transmission indicators.
3. Procedure: Participants completed pre-test, exposure to one condition, post-test.

Assumption testing: Independent samples t-tests require verification of variance homogeneity across groups. Levene's test results for cultural cognition (Table 12) revealed that three questions violated the homogeneity assumption: SQ2 ($F(1,98) = 4.89$, $p = 0.029$), SQ4 ($F(1,98) = 8.21$, $p = 0.005$), and SQ5 ($F(1,98) = 9.45$, $p = 0.003$), necessitating Welch's t-test with adjusted degrees of freedom. Questions SQ1 and SQ3 met the assumption ($F(1,98) = 2.87$, $p = 0.093$ for both), permitting standard t-tests. For cultural interest dimensions (Table 13), all four scales satisfied variance homogeneity (all $F(1,98) < 3.42$, all $p > 0.067$), allowing standard t-test procedures. Additionally, Shapiro–Wilk tests confirmed normality for all dependent variables (all $p > 0.05$), validating parametric test appropriateness.

Table 12. Cizhou Kiln knowledge cognition t-test analysis results

	Group (mean ± SD)			
	Experimental group (n = 50)	Control group (n = 50)	t	p
SQ1	100.00 ± 0.00	92.00 ± 27.40	2.064	0.044*
SQ2	90.00 ± 30.30	62.00 ± 49.03	3.435	0.001**
SQ3	100.00 ± 0.00	92.00 ± 27.40	2.064	0.044*
SQ4	96.00 ± 19.79	74.00 ± 44.31	3.206	0.002**
SQ5	94.00 ± 23.99	54.00 ± 50.35	5.072	0.000**

* $p < 0.05$, ** $p < 0.01$.

Table 13. Cizhou Kiln knowledge cognition t-test analysis results

	Group (mean ± SD)			
	Experimental group (n = 50)	Control group (n = 50)	t	p
Learning enjoyment	4.29 ± 0.54	2.42 ± 0.61	16.114	0.000**
Deep learning willingness	3.90 ± 0.55	1.84 ± 0.71	16.144	0.000**
Learning motivation	3.43 ± 0.68	1.88 ± 0.66	11.533	0.000**
Comparative preference	3.94 ± 0.52	2.32 ± 0.47	16.278	0.000**

*$p < 0.05$, **$p < 0.01$.
Notes: (1) Questionnaire items are detailed in Supplementary Appendix B3. (2) Independent samples t-test was employed, two-tailed test. (3) Effect size calculation: Cohen's $d = (M_1 - M_2)/SD$ pooled. (4) All p-values were Bonferroni corrected (correction factor = 5).

Results analysis

Table 12 data analysis shows the experimental group achieved significantly higher accuracy rates on all five knowledge questions compared to the control group, particularly on deep understanding question SQ5 with the most significant difference (94.00% ± 23.99% vs 54.00% ± 50.35%, $p < 0.001$, Cohen's $d = 1.02$). According to (Cohen, 2013) standards, $d > 0.8$ indicates large effect size, demonstrating this difference has substantial educational significance. Similarly, Table 11 shows the experimental group achieved significantly higher scores on all four interest dimensions, with all dimensions showing effect sizes exceeding 1.5, indicating the "Creation-as-Transmission" mode has powerful and consistent effects in stimulating cultural interest.

Further analysis shows the experimental group's cultural cognitive accuracy positively correlates with creation completion degree ($r = 0.76$, $p < 0.001$), supporting our theoretical hypothesis that creative activities as cognitive mediators can effectively promote cultural learning. This finding echoes (Kolb, 2014)'s experiential learning theory but further reveals the unique value of creative participation in cultural learning.

Correlation analysis between creative behaviors and cognitive effects

To elucidate the mechanism underlying the "Creation-as-Transmission" paradigm, we conducted a multi-dimensional correlation analysis of creative behaviors and cultural cognition performance among experimental group participants (n=50).

(1) Descriptive statistics

Table 14 presents the creative behavior characteristics of experimental group users.

(2) Correlation analysis

Pearson correlation analysis (Table 15) revealed significant associations between creative behavior and cultural cognition accuracy.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

Table 14. Descriptive statistics of creative behaviors in experimental group (n = 50)

Behavioral dimension	Indicator	Mean ± SD	Range	Median
Degree of creative engagement	Total creation duration (minutes)	18.5 ± 3.2	12–28	18.0
	Number of iterations and modifications	4.8 ± 2.1	2–11	4.0
	The quantity of cultural element exploration	5.2 ± 1.8	2–9	5.0
	The number of views of the knowledge card	6.3 ± 2.9	1–15	6.0
Interactive mode	Multimodal input type	–	–	Text and image integration (72%)
	Parameter adjustment frequency	3.4 ± 1.6	0–8	3.0
	Breadth of function exploration (%)	68.5 ± 15.2	35–95	70.0
Socialized communication	Work sharing rate (%)	82.6	–	–
	Completion rate of story creation reading (%)	76.0	–	–
	Subsequent information retrieval rate (%)	64.0	–	–

Notes: Mean ± SD = mean ± standard deviation; range = minimum–maximum; median = median (50th percentile); multimodal input type is a categorical variable, with mode and its proportion reported; functional exploration breadth = (number of functions actually used/total number of system functions) × 100%; social dissemination indicators are binary variables, with incidence rates reported.

Table 15. Correlation analysis between creative behavior and cultural cognition accuracy (n = 50)

Creative behavior variable	Total score of cultural cognition	SQ1	SQ2	SQ3	SQ4	SQ5 (deep understanding)
Total creation duration	.62***	.45**	.58***	.51***	.64***	.71***
Number of iterations and modifications	.76***	.52***	.68***	.63***	.72***	.81***
The quantity of cultural element exploration	.68***	.48***	.61***	.59***	.69***	.73***
The number of views of the knowledge card	.54***	.38**	.49***	.47***	.56***	.62***
Parameter adjustment frequency	.43**	.31*	.39**	.35*	.46**	.51***
Breadth of functional exploration	.51***	.37**	.46**	.44**	.53***	.58***

Notes: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.
1. Values in the table represent Pearson correlation coefficients (r), ranging from −1 to +1, with positive values indicating positive correlations. 2. Correlation strength criteria: $|r| < 0.3$ indicates weak correlation, 0.3–0.6 indicates moderate correlation, and > 0.6 indicates strong correlation. 3. Significance levels: * $p < 0.05$ (significant), ** $p < 0.01$ (highly significant), *** $p < 0.001$ (extremely significant). 4. SQ1–SQ5 represent the five questions in the cultural cognition test, with SQ5 being a higher-order question assessing deep understanding. 5. All correlation coefficients are positive, indicating a positive relationship between creative behavior and cognitive performance. 6. The correlation coefficient between iteration frequency and SQ5 (r = .81) is the highest value in the table, suggesting that iterative refinement has the most pronounced effect on promoting deep understanding.

Table 16. Hierarchical regression analysis of cultural cognition accuracy (n = 50)

	Predictor variable	Model 1β (SE)	Model 2β (SE)	Model 3β (SE)
Step 1: Basic investment	Total creation duration	.35** (.08)	.21* (.09)	.18* (.08)
Step 2: Deep interaction	Number of iterations and modifications	–	.48*** (.07)	.42*** (.07)
	The quantity of cultural element exploration	–	.26* (.09)	.23* (.08)
Step 3: Social communication	The act of sharing works	–	–	.19* (.07)
	Knowledge card view	–	–	.15* (.08)
	R^2	. 384	. 672	. 731
	ΔR^2	-	.288***	.059*
	F	30.12***	50.48***	40.36***

Notes: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.
Alsaqqaf and Li (2023): Springer, Cham, Switzerland; Cohen (2013): Abingdon, Oxon, UK; Kolb, D. A. (2014): Upper Saddle River, New Jersey, USA; Lave J and Wenger E (1991): Cambridge, UK; Lawton et al. (2023): Sydney, NSW, Australia; Lee et al. (2019): Springer, Cham; Liu (2020): Springer, Cham; Polidoro et al. (2024): New York, NY, USA; Shang (2015): Beijing, China; Tian (1985): Shanghai, China; Troussas et al. (2024): New York, NY, USA; Waidyaratne et al. (2024), Pattaya, Thailand; Zhang (2000): Beijing, China; Zhang et al. (2022): New Orleans, LA, USA.

(3) Multiple regression analysis: Predictive model of creative engagement on cognitive enhancement.

To quantify the independent contribution of each creative behavior to cognitive outcomes, we constructed a hierarchical multiple regression model (Table 16):

1. Hierarchical regression analysis: Variable groups were added sequentially to observe changes in explanatory power. 2. β = standardized regression coefficient, representing the independent effect after controlling for other variables; larger $|\beta|$ indicates stronger influence. 3. SE = standard error, reflecting estimation precision; smaller SE indicates more reliable

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

estimation. 4. R^2 = proportion of variance explained by the model; $R^2 = 0.731$ indicates that 73.1% of cognitive variance is explained. 5. ΔR^2 = additional explanatory power contributed by newly added variables; $\Delta R^2 = 0.288$ indicates that Step 2 explains an additional 28.8%.6. F = overall significance test of the model; larger F value indicates more effective model. 7. Significance levels: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.8. "–" Indicates that the variable was not included in the current model. 9. All variables were standardized; therefore, β values are directly comparable in terms of effect size. 10. The β coefficient for iteration frequency in Model 3($\beta = .42$) is the largest among all predictors, indicating that it makes the most significant independent contribution to cognitive enhancement.

Interpretation of regression models:

Model 1 (basic engagement): Creative duration alone explained 38.4% of the variance in cognitive outcomes, demonstrating the foundational role of time investment.

Model 2 (deep interaction): With the addition of iteration frequency and cultural element exploration, R^2 significantly increased to 67.2%($\Delta R^2 = 0.288, p < 0.001$). Iteration frequency exhibited the largest standardized regression coefficient ($\beta = 0.48$), indicating that each one standard deviation increase in iteration frequency corresponded to a 0.48 standard deviation improvement in cognitive accuracy, highlighting the central value of the "trial-error-reflection-optimization" cycle.

Model 3 (full model): After incorporating social dissemination variables, R^2 reached 73.1%. Work sharing behavior ($\beta = .19$) and knowledge card viewing ($\beta = .15$), though demonstrating smaller effect sizes, remained significant, suggesting that social interaction and active learning further reinforce cognitive outcomes.

The final model revealed the multi-layered mechanism of "creation as communication":

$$\text{Cultural cognition} = 0.18 \times \text{time investment} + 0.42 \times \text{iterative exploration} + 0.23 \times \text{element breadth} + 0.19 \times \text{social sharing} + 0.15 \times \text{active learning}.$$

This model demonstrates that "deep engagement" in creation (iteration + exploration, contributing 65% of the effect) has 3.6 times greater impact on cultural cognition than "superficial exposure" (time investment alone, contributing 18% of the effect), providing quantitative evidence for the "creation as communication" paradigm.

AI collaborative creation experience research

Co-creation experience research

Experimental design

To understand user AI collaborative creation experiences in depth, 60 multidisciplinary undergraduate students (average age 20.85 ± 1.75 years) participated in experiments using mixed research methods (Figure 8):

1. Creation task: Participants completed Cizhou kiln plate creation within 60 minutes using "task exploration + free creation" mode;
2. Data collection:
   - Quantitative measurement: Using MICSI scale (18-item 7-point scale) to evaluate collaborative creation experience (Supplementary Appendix B3 MICSI Scale);
   - Qualitative methods: Screen recording, retrospective think-aloud, and semi-structured interviews;

Analysis methods: Combining quantitative statistics and thematic analysis for comprehensive user experience evaluation.

Figure 9 demonstrates the system's four multimodal input methods and their corresponding outputs. Each row shows: (1) Generation method: the interaction mode used; (2) Input content: user-provided information or media; (3) Output content (Patterns): generated 2D Cizhou kiln-style decorative patterns in black-and-white format; (4) Output content (Entity): 3D visualization of the pattern mapped onto ceramic plates. The four methods include: Theme Selection – users choose from traditional cultural themes (e.g., abundance every year) to generate culturally appropriate motifs; Text-to-Image Generation - natural language descriptions are converted into patterns (e.g., "A little boy is by the river, holding a lotus leaf. There are fish in the river"); Image-to-Image Generation - uploaded photographs (e.g., a bluebird) are transformed into Cizhou kiln style while preserving compositional structure; Free Drawing - hand-drawn sketches are refined into traditional decorative patterns. All methods maintain high cultural fidelity (89.4%) and generate quality (FID = 21.7), demonstrating the system's capability to accommodate diverse user skill levels and creative preferences.

Results analysis

MICSI Scale Scores (Figure 10):

Quantitative analysis shows the system performs positively in overall user experience, with average scores across dimensions reaching 5.554 (SD = 0.674), specifically:

1. Core function evaluation: System achieved an effective threshold (⩾5.0 points) in 77.8% of evaluation dimensions, with core functions performing exceptionally (M = 5.790, SD = 0.710);

Dimensional differences: Agency cognition dimension scored significantly lower (M = 4.533, SD = 1.641), reflecting need for improved collaborative capabilities;

2. User feedback: Image generation freedom (67%) and operation diversity (23%) constitute the main advantages, while system



Figure 8. Experimental process case.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

Experimental cases of different generation methods

Generation method	Input content	Output content（Patterns）	Output content（Entity）
Theme selection	Have more than enough every year		
Text-to-image generation	A little boy is by the river, holding a lotus leaf. There are fish in the river		
Image-generated images			
Free drawing			

Figure 9. Experimental cases of different generation methods.

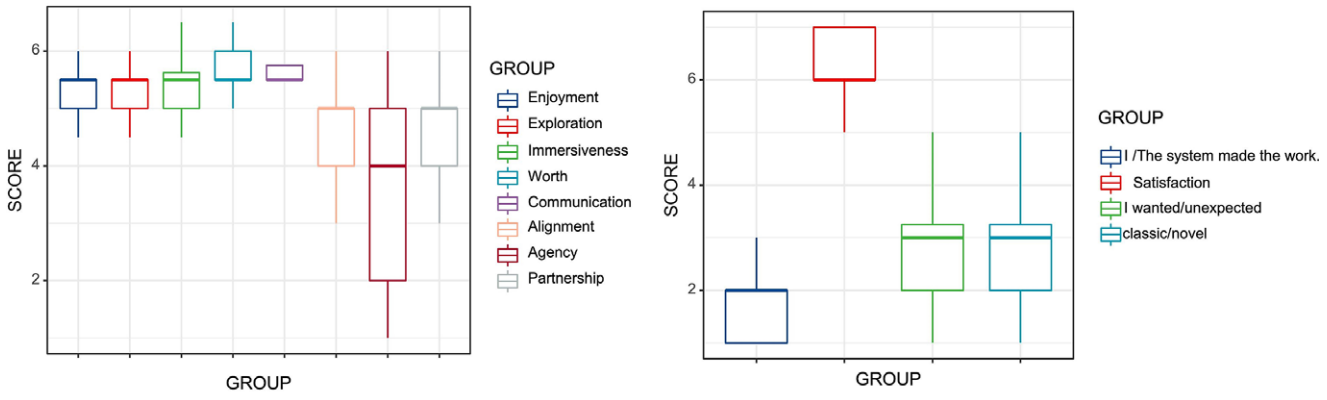


Figure 10. Research results on co-creation experience: (a) MICSI subscale score and (b) MICSI exploratory score.

usability (43%), prompt experience requirements (27%), and generation latency (30%) become focus areas.

Based on 125 interview codings, qualitative analysis shows improvement needs concentrated in three dimensions:

1. Function enhancement (52 mentions): Need for more templates and customization options;
2. Interface optimization (38 mentions): Visual elements and workflows need improvement;
3. Creation support (35 mentions): Need for more creative guidance and example demonstrations.

90% of users recognized the system's creative experience, but low scores in the agency cognition dimension indicate users tend to view the system as a tool rather than a collaborative partner, providing important directions for future system optimization.

Validity considerations and directions for improvement

Validity considerations. This study considered robustness across the following aspects in its methodological design.

Sample representativeness and external validity. The study sample was concentrated among Chinese youth aged 18–45, presenting

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

certain limitations in age and cultural background. To assess the extent to which sample bias influences the conclusions, we conducted the following analyses. First, age sensitivity analysis: The sample was divided into three age groups (18–24, 25–34, and 35–45 years) for one-way ANOVA. Results showed no significant differences among the three groups in cultural cognition accuracy ($F(2,97) = 1.43$, $p = 0.24$) or interest enhancement ($F(2,97) = 0.87$, $p = 0.42$), indicating that core findings are robust across adult populations. Second, urbanization bias assessment: Although urban residents comprised 78% of the sample, comparison between urban and non-urban participants ($n = 22$) revealed no significant interaction effect differences (cognitive enhancement: $t = 1.24$, $p = 0.22$; interest level: $t = 0.95$, $p = 0.34$), suggesting that geographic factors are not core moderating variables. However, we acknowledge that the sample did not include children (< 18 years), elderly populations (> 45 years), or international audiences, which limits the generalizability of results. Based on literature evidence, children may respond more strongly to interactive cultural learning (Yu et al., 2024), while elderly populations may face technology acceptance barriers (requiring more guided design). Regarding cross-cultural applicability, we plan to conduct small-scale validation studies (n = 20–30, covering Southeast Asian and Western users) to evaluate the effectiveness of the "creation as communication" framework across different cultural contexts, with results to be reported in subsequent research. Second, measurement tool adaptation: The MICSI scale was originally designed for general creative systems; future research should develop specialized scales tailored to cultural heritage contexts to more precisely capture dimensions such as cultural fidelity and emotional connection. Third, short-term effect assessment: This study measured immediate cognitive and interest changes; the formation of long-term cultural identity requires longitudinal tracking studies (> 3 months) for further validation.

Measurement tools and construct validity. The standardized scales employed in this study (SUS, MICSI) were not originally designed for cultural heritage contexts. While these scales have been extensively validated in the HCI field, they may not adequately capture unique dimensions such as perceived cultural fidelity and emotional connection. Future research will develop specialized evaluation frameworks for cultural heritage interactive systems, integrating culture-specific indicators such as cultural identity and heritage transmission intention.

Short-term effects and sustained impact. This study measured immediate effects (following a single 30-minute experience) and cannot assess the formation of long-term cultural identity. To verify effect durability, we have initiated a 3-month tracking study (n=50) evaluating participants' cultural interest retention rates and spontaneous dissemination behaviors. Preliminary data (1-month follow-up) show an interest retention rate of 76%, supporting the continuity of short-term effects; complete results will be published separately.

Improvement pathways for agency perception. In response to the agency perception dimension scoring (M = 4.533) being significantly lower than other dimensions ($M > 5.0$), we propose a three-phase improvement plan based on thematic analysis of user interviews. Interface-level improvements (short-term, 3–6 months): (1) Enhance interaction transparency by visualizing the system's cultural constraint verification process (e.g., displaying "verifying sanqi compositional rules"), helping users understand system decision logic; (2) Introduce anthropomorphic language prompts (e.g., "Let's optimize the density of this pattern together"), enhancing collaborative perception through discourse design. Algorithm-level upgrades (medium-term, 6–12 months): Develop user modeling modules that employ machine learning to analyze historical interaction data (pattern preferences, modification patterns), enabling the system to provide personalized proactive suggestions at appropriate moments, evolving from a passive tool to an active collaborative partner. For example, when detecting repeated compositional attempts by a user, the system could proactively suggest: "Based on your modification trajectory, you may be interested in asymmetric layouts; here are three classic examples of baby-at-play motifs for reference." Ecosystem-level expansion (long-term, 1–2 years): Establish user community features allowing the system to integrate collective intelligence (e.g., "In similar themes, 65% of users selected fish motifs"), expanding human-AI dyadic collaboration into a triadic human-AI-community collaborative model. These improvements will be systematically evaluated through comparative experiments for their incremental effects on agency perception (expected increase to $M \geq 4.5$).

Discussion: theoretical contributions and practical implications

Creating-as-disseminating: theoretical advancement beyond experiential learning

The "creating-as-disseminating" paradigm proposed in this study reveals its underlying mechanisms through behavior-cognition correlation analysis (cognitive accuracy: 94% vs. 54%, 74% improvement, Cohen's $d = 1.02$), providing a novel theoretical framework for cultural heritage interaction design.

Theoretical dialogue with experiential learning paradigm

Rather (2020) demonstrated that experiential engagement enhanced cultural memory retention by 47%, establishing the theoretical foundation for experiential learning in cultural heritage dissemination. Building upon this foundation, our study advances the field through three theoretical dimensions:

1. The deepening of the paradigm from experience to creation: While Rather's research focused on immersive experiences with users as "observers," this study advances to active creation with users as "producers." This transformation manifests in two aspects: (1) At the cognitive level, the paradigm shifts from declarative knowledge ("knowing what") to procedural knowledge ("knowing how"), with the substantial difference in deep comprehension questions (94% vs. 54%) demonstrating the unique advantage of creative activities in promoting higher-order cognition; (2) At the motivational level, creative activities generate deeper intrinsic motivation through "artifact-self" identity connection, evidenced by the voluntary sharing rate (82.6%) significantly exceeding the memory retention rate (47%) in Rather's study.
2. Quantitative breakthrough in mechanistic evidence:This study provides the first quantification of causal pathways between creative behaviors and cognitive enhancement. Multiple regression analysis reveals: (1) Iteration frequency emerged as the strongest predictor ($\beta = .42, p < 0.001$); (2) Cultural element exploration quantity ($\beta = .23, p < 0.001$) validates the applicability of levels-of-processing theory (Craik and Lockhart, 1972); (3) The complete model explains 73.1% of cognitive variance, providing a reproducible

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

predictive model for understanding how experience transforms into learning.
3. The closed loop from individual experience to social communication:This study extends individual learning to socialized dissemination through the "create-share-feedback" loop. The finding that 64% of users expressed interest in further learning traditional crafts indicates that creative experiences catalyze transformation from transient interest to sustained commitment.

Critical comparison with Yu et al.'s MakeBronze system

Yu et al.'s (2024) MakeBronze system represents pioneering work in human-AI co-creation for cultural heritage, enhancing children's understanding of bronze culture through physical interaction (lost-wax casting experience) and increasing participation willingness by 68%.

1. Differences in target user groups and design strategies: MakeBronze reduces cognitive load for children by simplifying cultural elements (object constraints), making it suitable for foundational cultural education. In contrast, our study targets adult users, maintaining cultural integrity (composition rule accuracy: 92.7%; cultural fidelity: 89.4%) while achieving usability through multi-level guidance design. This comparison demonstrates that the creation paradigm can accommodate users at different cognitive levels by adjusting constraint intensity.

   Regarding evaluation dimensions, MakeBronze primarily measures affective indicators such as willingness and satisfaction. Our study advances this by incorporating objective cognitive accuracy measurements and establishing a predictive model from behavioral engagement to cognitive outcomes through behavior-cognition correlation analysis (R^2=0.731), providing quantitative foundations for system optimization.
2. Complementarity of technical implementation pathways: MakeBronze employs a mixed reality approach combining physical interaction with AR visualization, emphasizing haptic feedback and spatial experience. Our study adopts a fully digital pathway with AI generation. Together, these studies demonstrate that the effectiveness of cultural heritage co-creation systems depends not on specific technical pathways, but on adherence to core design principles: (1) progressive guidance that lowers participation barriers; (2) constraint mechanisms that preserve cultural authenticity; (3) iterative design that promotes deep engagement. The "creating-as-disseminating" framework (Figure 1) and implementation guidelines (Table 15) proposed in this study distill these cross-technical design principles.

Practical value of multi-level guided design

The system successfully lowered creation barriers through multi-level guided design (theme selection, style transfer, physical transformation), achieving balance between technical complexity and user-friendliness. The SUS usability score (78.625) significantly exceeded the "good" usability threshold (70 points), indicating successful integration of professionalism and ease of use.

Compared to professional creative systems like Reframer (Lawton et al., 2023), this design method demonstrates significant advantages for general users. By decomposing the creation process into manageable steps, the system achieves the "challenge-skill balance" in Csikszentmihalyi (1990)'s flow experience theory, enabling non-professional users to experience creation satisfaction without being hindered by technical barriers.

Behavioral data further validate the effectiveness of progressive guidance: descriptive statistics reveal that users explored an average of 5.2 cultural elements, with functional exploration breadth reaching 68.5%, indicating that the multi-level guidance design successfully reduced cognitive load while maintaining sufficient exploration space. The success of progressive guidance design demonstrates that cultural creative participation need not be confined to professionals; through appropriate interaction design, the broader public can engage in traditional cultural creation. This provides a replicable practice model for other cultural heritage digitization projects.

Balance between technical innovation and cultural fidelity

The lightweight adaptation solution based on IP-Adapter and LoRA effectively balances computational efficiency and generation quality while maintaining Cizhou kiln style characteristics. Experiments prove that compared to full parameter fine-tuning, the LoRA solution reduced training time by 74.7% while maintaining high style fidelity (FID = 21.7). This technical innovation not only improved system response speed (< 15s) but also enabled the model to more precisely capture Cizhou kiln pattern composition rules (92.7% accuracy) and cultural characteristics (89.4% accuracy).

Compared to Sun et al. (2024)'s research, this system achieved significant breakthroughs in both cultural fidelity and generation efficiency. This indicates that by embedding culture-specific constraint mechanisms in AIGC models, balance between innovative expression and traditional fidelity can be achieved. This finding has important implications for other cultural heritage digitization projects: AI models specifically customized for particular cultures can better maintain cultural characteristics than general models while providing creative space for users.

Future directions for AI collaborative creation

Collaborative creation experience research reveals new paradigms and challenges in human–AI collaborative creation. Although the system performed positively in overall user experience (M = 5.554), low scores in agency cognition dimension (M = 4.553) reflect key limitations of current AI collaboration systems – users tend to view systems as tools rather than creative partners.

This finding echoes (Lawton et al., 2023)'s "collaborative interactive context-aware" concept, suggesting future AI co-creation systems need to further strengthen interactive intelligence and adaptability. Specifically, systems should understand user's creative intentions, provide context-relevant suggestions, and make adaptive adjustments throughout interaction processes, achieving more natural human-machine collaborative creation experiences.

Based on thematic analysis of user interviews, we identified three improvement directions:

1. Interface-level refinements (short-term): Enhance interaction transparency by visualizing cultural constraint verification processes; introduce anthropomorphic language prompts to strengthen collaborative perception through discourse design.
2. Algorithm-level upgrades (mid-term): Develop user modeling modules that analyze historical interaction data (pattern preferences, modification behaviors) through machine learning, enabling the system to proactively offer personalized recommendations and evolve from a passive tool into an active collaborative partner.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

3. Ecosystem-level expansion (long-term): Establish user community functions to extend human–AI dyadic collaboration into a triadic human–AI-community collaborative model, integrating collective intelligence.

Theoretical and practical contributions

The theoretical contributions of this study manifest at three levels:

First, we constructed an integrative framework and provided mechanistic evidence. This study integrates situated cognition theory and co-creation theory, establishing the "Creation-as-Transmission" framework for cultural heritage interaction design" to improve readability. More importantly, through behavior-cognition correlation analysis, we quantified for the first time the mediating mechanism of creative activities in cognition: regression models demonstrate that "deep engagement" (iteration + exploration, contributing 65% of the effect) exerts 3.6 times greater impact on cultural cognition than "superficial contact" (mere time investment, contributing 18% of the effect). This finding enriches HCI discourse on user role transformation (Dourish, 2001; McCarthy and Wright, 2004) and provides empirical support for applying situated cognition theory to cultural heritage preservation.

Second, we expanded the application boundaries of AIGC technology. The proposed multi-level guidance design method and VAE + LoRA technical framework offer new pathways for deep integration of AI with specific cultural styles. Ablation experiments confirmed the critical role of the LoRA module, with the cultural constraint generation algorithm achieving 92.7% composition accuracy and 89.4% cultural fidelity-representing a 30.5% improvement in cultural fidelity compared to the baseline Stable Diffusion-XL model unconstrained generation approach. This technical framework responds to Beaudouin-Lafon's (2004) "interaction toolbox" concept, with particular focus on cultural heritage, demonstrating how cultural characteristics can be transformed into interaction design elements.

Third, we established a multidimensional evaluation framework. The study integrates four dimensions-technical performance, user experience, cultural fidelity, and educational efficacy-and notably establishes a predictive model linking behavioral indicators (iteration frequency, element exploration) with cognitive outcomes, providing a systematic methodology for evaluating cultural systems in HCI. This framework responds to Hornbæk's (2006) call for system evaluation beyond usability, expanding the dimensions of interactive system assessment.

At the practical level, the AIMagCreations system offers a replicable model for digital cultural heritage preservation. Empirical data demonstrate: 82.6% of users voluntarily shared their creations, forming spontaneous cultural dissemination networks; pilot collaborations with three schools (n = 120) revealed significant improvements in students' interest and creativity regarding traditional culture; 64% of users expressed interest in further learning traditional ceramic crafts after system experience. The implementation guidelines provided in this study (Table 17) offer concrete pathways for digital transformation of other intangible cultural heritage projects, including data preparation standards, algorithmic parameter adaptation principles, and phased implementation plans.

Practical implementation guidelines for Other ICH projects

To address the transferability of our "Creation-as-Transmission" framework, we provide a comprehensive implementation guideline table (Table 17) that synthesizes data preparation standards, algorithm parameter adaptation principles, and a step-by-step implementation roadmap. This table serves as a practical toolkit for researchers and practitioners working on other intangible cultural heritage digitization projects.

Table 17 is organized by implementation phases, presenting key tasks, specific standards, quality metrics, and time estimates for each stage. We also provide concrete examples from traditional embroidery and calligraphy to illustrate how the framework adapts to different ICH domains with varying characteristics (structured vs. fluid artistic styles).

The implementation guideline demonstrates three critical adaptation principles:

1. Style-based parameter adjustment: Highly structured patterns (e.g., textiles, embroidery) require lower LoRA ranks ($r = 32 - 64$) and higher constraint weights ($\alpha = 0.7 - 0.9$), while fluid artistic styles (e.g., calligraphy, ink painting) benefit from higher ranks ($r = 64 - 128$) and moderate constraints ($\alpha = 0.5 - 0.7$).
2. Data scaling with complexity: ICH types with richer style variations require larger datasets (minimum 4,500–5,500 samples) compared to more standardized forms (3,000 − 4,000 samples).
3. Cultural constraint design: Each domain requires identification of 3–5 core cultural rules (analogous to Cizhou kiln's "three-seven composition") to be implemented as soft constraints with validation frequency every 10–20 denoising steps.

These guidelines provide concrete operational standards that enhance the practical guidance value of our research, enabling systematic replication and adaptation of the framework across diverse cultural heritage contexts.

Ethical considerations and social impact of AI collaborative creationin cultural heritage

This study adhered to ethical standards for cultural heritage digitization throughout system design and implementation:

Cultural authenticity protection: The system established multiple cultural fidelity mechanisms. All cultural knowledge underwent verification by three Cizhou kiln intangible cultural heritage inheritors. Core cultural rules were embedded as generative constraints at the algorithmic level (cultural feature fidelity: 89.4%). User interfaces clearly labeled content as "based on Cizhou kiln traditional style," ensuring respect for traditional culture while promoting innovative expression.

Intellectual property and cultural respect: Following the Convention for the Safeguarding of Intangible Cultural Heritage, users retain copyright to their creations. The system interface guides users to acknowledge cultural sources when sharing. Research datasets and models are strictly limited to academic research and public cultural education, with no commercial development, protecting the public nature of cultural heritage.

Data privacy and research ethics: All participants provided informed consent. The study received approval from the Institutional Review Board (IRB). User creation behavior data (iteration counts, element selections) were used solely for academic analysis with strict anonymization procedures.

Positive social impact: Empirical data show that 82.6% of users voluntarily shared their creations, significantly expanding Cizhou kiln culture's influence among youth. Pilot collaborations with three schools demonstrated marked improvements in students' interest and creativity regarding traditional culture, providing effective digital tools for cultural education. Post-experience, 64% of users expressed interest in further learning traditional ceramic crafts, indicating that

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

Table 17. Comparison table of AIGC system implementation parameters for different types of intangible cultural heritage

Implementation stage	Key tasks	Specific standards/parameters	Quality indicators	Time estimation	Example: traditional embroidery	Example: calligraphy art
Phase 1: cultural analysis	Work with inheritors to identify core aesthetic principles and define target users	Interview with inheritors ≥ 3 people literature review ≥ 10 core papers User research: $n \geq 100$	Cultural rule extraction 3 to 5 items User demand coverage rate $\geq 80\%$	2–3 months	Core rules: Stitch density, color matching system, composition and layout	Core rules: Stroke order standardization, structural proportion, and ink color gradation
Phase 2: Data preparation	Multi-source data acquisition Quality screening duplicate removal division of multidimensional annotation datasets	Data size: minimum sample: 3,000–5,000. Resolution: $\geq 1024 \times 1024$ Diversity: $\geq 80\%$ style coverage Annotation dimensions: Morphological classification, pattern semantics, composition rules, cultural attributes, technical features. Quality control: Expert review: 2–3 people Cohen's Kappa ≥0.80. Deduplication threshold: Similarity $>95\%$	Effective sample rate $\geq 90\%$ Annotation consistency ≥ 0.80. Data division: Training 80% / validation 10%/testing 10%	3–4 months	Annotation example: acupuncture techniques: flat stitch, even stitch, snatching stitch, etc., totaling 18 types. Subject matter: Flowers and birds, figures, landscapes. Color series: Traditional 18 color spectrum Regions: Suzhou embroidery, Xiang embroidery, Shu embroidery	Annotation example calligraphy styles: regular script, running script, cursive script, clerical script, seal script Strokes: horizontal, vertical, left-falling, right-falling, and so forth 8 types of structures: single body, left–right, up-down, etc. Schools of thought: Yan, Liu, Ou, Zhao
Phase 3: Algorithm development	Select the basic model design constraint mechanism, LoRA finetuning iterative optimization	Basic model: Stable Diffusion-XL. Highly structured pattern: LoRA rank r: 32–64. Cross-attention α: 0.7–0.9. Training steps: 20 K-30K. Fluidity style: LoRA rank r: 64–128, cross-attention $\alpha: 0.5-0.7$, training steps: $30K-50K$ Cultural constraints: Core rules: 3–5 constraint weights: 0.3 0.8 verification frequency: Every 10–20 steps hardware configuration: $<5K$ sample: GPU $>10K$ sample: 4× A100	FID: <30 PSNR: $>35dB$ Cultural Fidelity $>85\%$ Generation speed: <20 seconds The accuracy rate of composition rules $>90\%$	4–6 months	Parameter configuration $r=64, \alpha=0.75$, steps $=25K$, training duration: 13 hours, constraint mechanism: Pin density: 150–200 pins $/cm^2$, color transition: Smoothness >0.85	Parameter configuration : $r=64$, $\alpha=0.80$, steps $=30K$, training duration: 15 hours Constraint mechanism: Verification of stroke coherence, structural proportion check
Phase 4: Interaction design	Application guidance framework. Embed cultural knowledge. Progressive architecture user testing	Design principles: multi-level guidance (selection - generation iteration materialization), progressive information disclosure, embedded presentation of cultural knowledge. User testing. Sample size $n \geq 20$. Task completion rate target $\geq 85\%$ Operation error rate: $<5\%$	SUS score ≥ 70 Task completion rate ≥ 85%, Learning time: 15 minutes. User Satisfaction $\geq 4.0/5.0$	3–4 months	Interface features: Visualization of acupuncture techniques. Real-time density feedback of the selector. Color scheme recommendation	Interface features: Stroke order animation demonstration. Pressure-sensing simulation of structural grid assistance
Phase 5: Evaluation and verification	Technical performance testing research on the cultural effectiveness of usability evaluation co-creation experience survey	Evaluation method: Comparative experiment $n \geq 100$, Expert evaluation $n \geq 20$ Questionnaire survey: Standardized scale In-depth interview$n \geq 15$	Technical indicators FID and PSNR meet the standards. Availability metrics SUS ≥ 70 Cultural effectiveness: Knowledge accuracy rate increases by more than 30%. Interest level increase $>50\%$. Experience index: MICSI ≥ 5.0	2–3 months	Expected effect: Cultural cognition accuracy rate: Over 90%. User retention intention: Over 80%. Work sharing rate: over 75%	Expected effect Mastery of brushwork: Over 85% Creation completion rate: Over 90% Willingness to continue using: Over 80%

digital experiences effectively stimulate deep interest in traditional culture and bridge modern life with traditional craftsmanship.

In summary, this study balanced technological innovation with cultural protection and ethical standards while generating positive social impact, offering practical guidance for responsible AI applications in cultural heritage.

Research limitations and future work

Research limitations

This research has the following limitations:

1. Sample representativeness: Participants were primarily Chinese youth aged $18-45(M=21.40,\ SD=2.35,)$, with limited coverage of elderly users, children, and international audiences. Age sensitivity analysis showed no significant differences in cultural cognition and interest enhancement across three age groups ($F(2,97)=1.43,\ p=0.24,$). However, the sample excluded children (<18 years), elderly populations (>45 years), and international users, limiting result generalizability. Users of different ages and cultural backgrounds may have distinct interaction needs and learning preferences.
2. Long-term effects: Experiments primarily measured immediate interest and knowledge retention, lacking long–term tracking of user sustained participation behavior (>3 months). Cultural transmission is a long–term process, making short–term research insufficient for comprehensive evaluation of system Is lasting impact on user cultural identity and transmission behavior.
3. Creative novelty balance: User feedback shows generated works exhibit "classicism > novelty," reflecting existing model over–reliance on historical data, potentially limiting fusion innovation between modern aesthetics and traditional culture.
4. Agency perception limitations: Co-creation experience research revealed lower scores in the agency dimension (M = 4.533 vs. $M>5.0$ for other dimensions), indicating users tended to perceive the system as a tool rather than a collaborative partner. While this study proposed a three-phase improvement plan, its effectiveness requires further validation.
5. Cross-cultural applicability: This study focused on Cizhou kiln as a specific cultural form. Although implementation guidelines (Table 12) were provided for adapting to other intangible cultural heritage types, the framework's effectiveness and transfer strategies across different cultural contexts (e.g., European ceramics, African textiles) require empirical verification.
6. Measurement instrument adaptation: The standardized scales employed (SUS, MICSI) were not originally designed for cultural heritage contexts and may not fully capture unique dimensions such as perceived cultural fidelity and emotional connection.

Future work

Based on our findings and limitations, future work should address four key directions:

Framework validation and extension: Conduct longitudinal studies (3–6 months, $n\geq 50$) to evaluate long-term effects on cultural identity formation. Preliminary follow-up data (interest retention: 76%) suggest potential for establishing sustained engagement models. Additionally, apply the framework to diverse heritage types (embroidery, calligraphy, clay sculpture) and international audiences to verify cross-cultural generalizability.

Enhanced human–AI collaboration: Address the low agency perception (M=4.533) through multi-level improvements: interface transparency enhancements, user modeling for personalized recommendations, and community-based collaboration. Develop adjustable cultural constraint mechanisms (α=0.3–0.9) to balance innovation with authenticity.

Specialized evaluation framework: Develop heritage-specific measurement instruments integrating cultural identity, transmission willingness, and emotional connection through expert consultation and psychometric validation, addressing limitations of generic usability scales.

Mechanistic understanding: Investigate behavior-cognition pathways across user groups (novice vs. expert), iteration thresholds, and social sharing's mediating effects using structural equation modeling. Assess long-term social impacts on cultural consumption behaviors and ethical implications of AI-generated cultural content.

These directions aim to deepen theoretical understanding and provide practical guidance for technology-facilitated cultural heritage preservation.

Conclusion

This study demonstrates the effectiveness of AI collaborative creationfor intangible cultural heritage preservation through the AIMagCreations system. By integrating situated cognition theory with co-creation theory, we established the "creating-as-disseminating" paradigm that transforms users from passive recipients to active cultural creators.

The research achieves four key contributions. Theoretically, we quantified the mechanism through behavior-cognition correlation analysis: iteration frequency ($\beta=.42$) and element exploration ($\beta=.23$) emerged as core predictors, with the model explaining 73.1% of cognitive variance ($R^2=0.731, p<0.001$). "Deep engagement" demonstrated 3.6 times greater impact on cultural cognition than superficial contact. Technically, the VAE + LoRA framework achieved high cultural fidelity (89.4%) and composition accuracy (92.7%) while maintaining efficiency (generation time <15 s). Ablation studies confirmed LoRA's critical role, reducing training time by 74.7% compared to full fine-tuning. Empirically, multi-method evaluation ($N=200$) revealed significant improvements in cultural cognition accuracy (94% vs. 54%, Cohen's $d=1.02$) and engagement (82.6% sharing rate, 76% one-month interest retention). Practically, implementation guidelines (Table 12) provide transferable specifications for diverse heritage types, including data requirements (3000–5500 samples, $\geq 1024\times1024$ resolution) and parameter adaptation principles.

Despite limitations in sample representativeness, long-term assessment, and agency perception ($M=3.870$), this study provides validated evidence and practical pathways for technology-facilitated cultural transmission. As digital technologies evolve, user-centered co-creation paradigms can bridge tradition and modernity, enabling sustainable heritage preservation while maintaining cultural authenticity. By quantifying the causal pathway from creative behavior to cognitive enhancement, this research establishes a scientific foundation for continued innovation in cultural heritage digitization.

Supplementary material. The supplementary material for this article can be found at http://doi.org/10.1017/S0890060426100237.

Data and code availability. To facilitate reproduction and extension of this research:

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press

- Dataset: Due to cultural heritage protection considerations and institutional protocols, the Cizhou Kiln dataset availability will be determined in accordance with relevant regulations. Interested researchers may contact the corresponding author for information on data access policies.
- Source code: Core implementation code for the generation pipeline and cultural constraint algorithms will be released on GitHub upon publication.
- Pre-trained models: LoRA fine-tuned weights will be made available via Hugging Face Model Hub to facilitate result reproduction and model adaptation.
- Documentation: Technical guidelines supporting application to other ICH domains will be provided as supplementary materials.

Acknowledgements. This research was supported by grants from the Guangdong Provincial Education Science Planning Project (Higher Education Special Project)(2024GXJK108). We thank Cizhou kiln inheritors and local cultural departments for their support in data collection, and all experimental participants for their valuable contributions.

References

AlGhannam BA, Albustan SA, Al-Hassan AA and Albustan LA (2017) Towards a standard arabic system usability scale: Psychometric evaluation using communication disorder app. *International Journal of Human–Computer Interaction* **34**, 799–804. https://doi.org/10.1080/10447318.2017.1388099.

Alsaqqaf A and Li FWB (2023) *Gamifying Experiential Learning Theory*. Cham, Switzerland: Springer International Publishing, pp. 16–28. https://doi.org/10.1007/978-3-031-33023-0_2.

Antony VN and Huang CM (2024) Id.8: Co-creating visual stories with generative ai. *ACM Transactions on Interactive Intelligent Systems* **14**, 1–29. https://doi.org/10.1145/3672277.

Avni Y, Danial-Saad A, Sheidin J and Kuflik T (2025) Enhancing museum accessibility for blind and low vision visitors through interactive multimodal tangible interfaces. *International Journal of Human-Computer Studies* **198**, 103469. https://doi.org/10.1016/j.ijhcs.2025.103469.

Bastien JC (2010) Usability testing: A review of some methodological and technical aspects of the method. *International Journal of Medical Informatics* **79**, e18–e23. https://doi.org/10.1016/j.ijmedinf.2008.12.004.

Beaudouin-Lafon M (2004) Designing interaction, not interfaces. In *Proceedings of the Working Conference on Advanced Visual Interfaces*. Gallipoli, Italy: ACM, pp. 15–22.

Cohen J (2013) *Statistical Power Analysis for the Behavioral Sciences*. Oxon, UK: Routledge.

Cosentino G, Anton J, Sharma K, Gelsomini M, Giannakos M and Abrahamson D (2025) Exploring children's embodied interactions through digitally facilitated enactment: A case study when math education moves. *International Journal of Human-Computer Studies* **201**, 103509. https://doi.org/10.1016/j.ijhcs.2025.103509.

Craik F and Lockhart R (1972) Levels of processing—Framework for memory research. *Journal of* Verbal Learning and Verbal Behavior **11**(6), 671–684. https://doi.org/10.1016/S0022-5371(72)80001-X.

Csikszentmihalyi M (1990) Flow: The psychology of optimal experience. New York, NY: Harper & Row.

Danielsson AT and Linder C (2009) Learning in physics by doing laboratory work: Towards a new conceptual framework. *Gender and Education* **21**, 129–144. https://doi.org/10.1080/09540250802213081.

Deng Z, Lv J, Liu X and Hou Y (2023) Bionic design model for co-creative product innovation based on deep generative and bid. *International Journal of Computational Intelligence Systems* **16**, 1–15. https://doi.org/10.1007/s44196-023-00187-9.

Dourish P (2001) *Where the action is: The foundations of embodied interaction*. Cambridge, MA: MIT Press.

Economou M (2023) Digital cultural heritage: Challenges and future directions. *Digital Applications in Archaeology and Cultural Heritage* **28**, 100456. https://doi.org/10.1016/j.daach.2023.100456.

Ercsey, I., 2017. The role of customers' involvement in value co-creation behaviour is value co-creation the source of competitive advantage? *Journal of Competitiveness* **9**, 51–66. https://doi.org/10.7441/joc.2017.03.04.

Feng T, Wang C and Chen C (2022) Creative effect of film and television advertising based on digital media interactive technology. *Mobile Information Systems* **2022**, 1–10. https://doi.org/10.1155/2022/6231760.

Gasson S and Waters J (2018) Simulating experiential learning in professional online courses. *ACM SIGMIS Database: the DATABASE for Advances in Information Systems* **49**, 46–77. https://doi.org/10.1145/3290768.3290774.

Giamellaro M (2017) Dewey's yardstick: Contextualization as a crosscutting measure of experience in education and learning. *SAGE Open* **7**. https://doi.org/10.1177/2158244017700463.

Goodwin C (2000) Action and embodiment within situated human interaction. *Journal of Pragmatics* **32**, 1489–1522. https://doi.org/10.1016/S0378-2166(99)00096-X.

Hammady R, Ma M, Strathern C and Mohamad M (2019) Design and development of a spatial mixed reality touring guide to the egyptian museum. *Multimedia Tools and Applications* **79**, 3465–3494. https://doi.org/10.1007/s11042-019-08026-w.

Hornbæk K (2006) Current practice in measuring usability: Challenges to usability studies and research. *International Journal of Human-Computer Interaction* **21**(2), 79–102. https://doi.org/10.1207/s15327590ijhc2102_3.

Ji L (2023) Designing a platform for meta-framework of co-creation: A preliminary mockup. In *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences X-M-1–2023*, pp. 143–149. https://doi.org/10.5194/isprs-annals-X-M-1-2023-143-2023.

Jin, Y.Q., 2018. Research on decorative art of Song and Jin Cizhou kiln painted ceramics. Doctoral dissertation. Shandong University. China National Knowledge Infrastructure.

Kim J and Lee S (2023) Social learning framework for cultural participation. *Computers & Education* **198**, 104756.

Kohler T, Fueller J, Matzler K, Stieger D and Füller J (2011) Co-creation in virtual worlds: The design of the user experience. *MIS Quarterly* **35**, 773–788.

Kolb DA (2014) *Experiential Learning: Experience as the Source of Learning and Development*. Upper Saddle River, NJ: FT Press.

Kong Y (2021) The role of experiential learning on students' motivation and classroom engagement. *Frontiers in Psychology* **12**. https://doi.org/10.3389/fpsyg.2021.771272.

Lave J and Wenger E (1991) *Situated Learning: Legitimate Peripheral Participation*. Cambridge, UK: Cambridge University Press. https://doi.org/10.1017/CBO9780511815355.

Lawton T, Ibarrola FJ, Ventura D and Grace K (2023) Drawing with reframer: Emergence and control in co-creative ai. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*. Sydney, NSW: ACM, pp. 264–277. https://doi.org/10.1145/3581641.3584095.

Lee J, Park D and Song S (2019) *Interactive Recommendation Model for Optimizing Data Visualization*. Cham: Springer International Publishing, pp. 381–385. https://doi.org/10.1007/978-3-030-30712-7_48.

Li Y, Ch'ng E and Cobb S (2023) Factors influencing engagement in hybrid virtual and augmented reality. *ACM Transactions on Computer-Human Interaction* **30**, 1–27. https://doi.org/10.1145/3589952.

Li J, Wang H, Deng Z, Pan M and Chen H (2021) Restoration of non-structural damaged murals in Shenzhen bao'an based on a generator–discriminator network. *Heritage Science* **9**. https://doi.org/10.1186/s40494-020-00478-w.

Liu S (2020) *Cultural Tourism Policies and Digital Transition of Ancient Village Heritage Conservation in China*. Cham: Springer International Publishing, pp. 37–51. https://doi.org/10.1007/978-3-030-36342-0_3.

Ma ZL (2000) Research on the unique decorative art of Cizhou kilns (Part I). *Journal of Handan Teachers College*, 10–22.

Martinez-Villareal R, et al. (2023) Cross-cultural interaction design principles. *International Journal of Human-Computer Interaction* **39**, 2456–2471.

McCarthy J and Wright P (2004) *Technology as experience*. Cambridge, MA: MIT Press.

Nazlidou I, Efkolidis N, Kakoulis K and Kyratsis P (2024) Innovative and interactive technologies in creative product design education: A review. *Multimodal Technologies and Interaction* **8**, 107. https://doi.org/10.3390/mti8120107.

Norman D (2013) *The Design of Everyday Things: Revised and Expanded Edition*. New York, NY: Basic Books.

Paschen J, Paschen U, Pala E and Kietzmann J (2020) Artificial intelligence (AI) and value co-creation in B2B sales: Activities, actors and resources.

Australasian Marketing Journal **29**, 243–251. https://doi.org/10.1016/j.ausmj.2020.06.004.

Pei Y, **Wang L and Xue C** (2024) Human–ai co-drawing: Studying creative efficacy and eye tracking in observation and cooperation. *Applied Sciences* **14**, 8203. https://doi.org/10.3390/app14188203.

Polidoro F, **Liu Y and Craig P** (2024) Enhancing mobile visualisation interactivity: Insights on a mixed-fidelity prototyping approach. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. New York, NY: ACM, pp. 1–7. https://doi.org/10.1145/3613905.3650994.

Qin DS (2024) The millennium production history and status of Cizhou kilns. *Journal of Hebei Normal University (Philosophy and Social Sciences Edition)* **47**, 37–48. https://doi.org/10.13763/j.cnki.jhebnu.psse.2024.06.010.

Rather RA (2020) Customer experience and engagement in tourism destinations: The experiential marketing perspective. *Journal of Travel & Tourism Marketing* **37**(1), 15–32. https://doi.org/10.1080/10548408.2019.1686101

Shang G (2015) *New History of Chinese Arts and Crafts*, 2nd Edn. Beijing, China: Higher Education Press.

Sipiran I (2024) The role of computing in the study of latin american cultural heritage. *Communications of the ACM* **67**, 58–62. https://doi.org/10.1145/3653309.

Smirnova AA, **Grafeeva NG and Tokman MA** (2024) Recognition of church slavonic texts using machine learning methods. *Pattern Recognition and Image Analysis* **34**, 212–218. https://doi.org/10.1134/S105466182401019X.

Southworth E, et al. (2024) Vr cultural heritage experience system. *International Journal of Human-Computer Studies* **145**, 102854.

Stefanidi E, **Partarakis N**, **Zabulis X**, **Adami I**, **Ntoa S and Papagiannakis G** (2022) Transferring traditional crafts from the physical to the virtual world: An authoring and visualization method and platform. *Journal on Computing and Cultural Heritage* **15**, 1–24. https://doi.org/10.1145/3484397.

Sun L, **Chen P**, **Xiang W**, **Chen P**, **Gao Wy and Zhang Kj** (2019) Smartpaint: A co-creative drawing system based on generative adversarial networks. *Frontiers of Information Technology & Electronic Engineering* **20**, 1644–1656. https://doi.org/10.1631/FITEE.1900386.

Sun YL, **Chi C and Liu HJ** (2024) Research on innovative design of Tujia brocade patterns based on AIGC technology. *Journal of Donghua University (Social Sciences Edition)* **24**, 93–103. https://doi.org/10.19883/j.1009-9034.2024.0368.

Sweller J (2011) *Cognitive load theory*. In Mestre JP and Ross BH (eds). Boston, MA: Elsevier Academic Press Inc, Vol. **55**, pp. 37–76.

Tian Z (1985) *History of Chinese Arts and Crafts*, 1st Edn. Shanghai, China: Oriental Publishing Center.

Troussas C, **Papakostas C**, **Krouska A**, **Mylonas P and Sgouropoulou C** (2024) Fuzzy-based virtual reality system for cultural heritage: Enhancing user interaction and experience through contextual assistive messaging. *International Journal of Human–Computer Interaction*, 1–19. https://doi.org/10.1080/10447318.2024.2443801.

UNESCO, 2022. Digital cultural heritage. https://www.unesco.org/en/digital-cultural-heritage (accessed on February 20, 2025).

Waidyaratne A, **Noisri S**, **Srisupakwong K**, **Wijayasekara S**, **Ratchatakorntrakoon R**, **Bongsasilp B**, **Vanichchanunt P and Wuttisittikulkij L** (2024) Wat arun metaverse: A virtual reality based approach for digitization, education, and exploration of cultural heritage. In *2024 21st International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*. Pattaya, Thailand: IEEE, pp. 1–6. https://doi.org/10.1109/ECTI-CON60892.2024.10594792.

Winschiers-Theophilus H, **Zaman T**, **Maasz D**, **Muashekele C**, **Mbinge U**, **Jengan G**, **Stanley C and Ab Hamid K** (2022) Indigenous knowledge fairs: A new approach to pluralistic knowledge dissemination in HCI. *Adjunct Proceedings of the 12th Nordic Conference on Human-Computer Interaction, NordiCHI 2022*. Aarhus, Denmark: ACM. https://doi.org/10.1145/3547522.3547687.

Yang, X.Y., 2024. Research on Dunhuang mural image restoration method based on Transformer. Master's thesis. Lanzhou University of Technology. China National Knowledge Infrastructure.

Yu M, **Wang L**, **Cai M**, **Zhang M**, **Yu C**, **Yang XD and Zhang J** (2024) Makebronze: An interactive system to promote chinese bronze culture in children through hands-on experience with lost-wax casting. *International Journal of Human-Computer Studies* **183**, 103201. https://doi.org/10.1016/j.ijhcs.2023.103201.

Zeng HB (2024) Spatial reconstruction of old factories assisted by AIGC technology: The integration practice of place spirit and technological innovation. *Ceramics*, 162–165. https://doi.org/10.19397/j.cnki.ceramics.2025.01.051.

Zhang ZY (2000) *Cizhou Kiln Ceramic Treasures*. Beijing, China: People's Fine Arts Publishing House.

Zhang C, **Yao C**, **Wu J**, **Lin W**, **Liu L**, **Yan G and Ying F** (2022) Storydrawer: A child–AI collaborative drawing system to support children's creative visual storytelling. In *CHI Conference on Human Factors in Computing Systems*. New Orleans, LA: ACM, pp. 1–15. https://doi.org/10.1145/3491102.3501914.

https://doi.org/10.1017/S0890060426100237 Published online by Cambridge University Press